

The galactic acceleration scale and the cosmological constant: the coefficient on SPARC, and what it takes to measure its redshift evolution

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ABSTRACT

The acceleration scale a_0 of the radial acceleration relation is close to $c\sqrt{G\rho_\Lambda}$, the only acceleration built from c , G and the density of the cosmological constant. We write $a_0 = \kappa c\sqrt{G\rho_\Lambda}$, measure κ and ask what testing its redshift evolution requires. On SPARC, with distances and ρ_Λ on one H_0 , three estimators give $\kappa = 0.45 \pm 0.07$, 0.58 ± 0.18 and 0.43 ± 0.08 . The value $1/2$ is consistent with all three, as are Milgrom’s $cH_\Lambda/2\pi$ and $cH_0/2\pi$; a published joint inference puts $1/2$ at 2.6σ . κ is fitted, degenerate with H_0 and set by the stellar mass-to-light zero point, with a 9.5 per cent calibration floor. A Λ -anchored scale is constant; a scale tied to the critical density grows as $H(z)$ (+0.58 dex at $z = 2.5$); the emergent scale of Λ CDM haloes rises by +0.22 dex for haloes of discs observable at low acceleration. Existing samples decide nothing: a 0.05 dex per unit redshift drift in the baryonic-mass calibration erases the trend of 99 rotation curves at $0.6 < z < 2.5$, the MUSE-DARK rise at $z \approx 1$ is not recovered with SED stellar masses plus molecular gas, and a lean towards $a_0 \propto H(z)$ at $z \approx 1.5$ is not robust. Inferring a_0 amplifies errors unless $g_{\text{bar}} < 0.3a_0$. Expected odds of 20:1 need about eight lensed discs at $z \approx 2.5$ measured to 0.20 dex, twenty for a 90 per cent chance of reaching 20:1, and a common baryonic-mass calibration good to 0.06 dex, which no gas tracer yet provides.

Key words: gravitation – galaxies: kinematics and dynamics – galaxies: high-redshift – cosmological parameters – dark energy – dark matter

1 INTRODUCTION

In rotationally supported galaxies the observed centripetal acceleration, $g_{\text{obs}} = V^2/R$, is a function of the acceleration g_{bar} that the visible stars and gas would produce on their own. This radial acceleration relation (McGaugh et al. 2016; Lelli et al. 2017) contains one dimensional constant, $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$. Above a_0 the two accelerations agree; far below it, $g_{\text{obs}} \approx \sqrt{g_{\text{bar}} a_0}$, which is also the content of the baryonic Tully–Fisher relation (McGaugh 2012; Lelli et al. 2019). The same constant was introduced by Milgrom (1983) as the scale of a modified dynamics (MOND), and he noted at once that a_0 is of the order of cH_0 . In their review, Famaey & McGaugh (2012) describe a_0 as of the order of the square root of the cosmological constant in natural units, and Milgrom (2020) writes the coincidence in two forms, $2\pi a_0 \approx cH_0$ and $2\pi a_0 \approx c^2\sqrt{\Lambda/3}$.

Those two forms are different physical statements. If a_0 follows the expansion rate it was larger in the past, by a factor 3.8 at $z = 2.5$. If it follows the cosmological constant it never changes. Today the two differ only by the factor $\Omega_\Lambda^{-1/2} = 1.21$, which is inside the uncertainty of any measurement of a_0 , so local data cannot tell them apart. A third possibility is that there is no fundamental a_0 at all: in Λ CDM the relation is an emergent property of galaxies in dark-matter haloes (Navarro et al. 2017; Ludlow et al. 2017; Keller & Wadsley 2017), and the characteristic acceleration of haloes also grows with redshift.

This paper takes the Λ reading as a hypothesis and asks two em-

pirical questions. We write

$$a_0 = \kappa c\sqrt{G\rho_\Lambda}, \quad (1)$$

where ρ_Λ is the mass density equivalent to the cosmological constant and κ is a pure number, and we ask:

- (i) *What is κ on present data, and how well can it be known?* (Section 3, which also states its degeneracy with the Hubble constant.)
- (ii) *What would it take to measure whether a_0 evolves, and so to separate equation (1) from the alternatives?* (Section 4.)

Section 2 sets out the relation, Section 5 lists what it does not explain, and Section 6 concludes.

The idea has a long history and we claim no priority for it. Milgrom (1999) obtained MOND-like behaviour from the Unruh temperature of an accelerated observer in de Sitter space, with a coefficient corresponding to $a_0 = 2cH_\Lambda$. Limbach et al. (2008) performed the redshift test itself: they confronted the cH_0 - and ρ_Λ -coupled forms of $a_0(z)$ with two Tully–Fisher samples reaching $z = 1.2$, found both couplings excluded within the formal uncertainties and, once systematic uncertainties were allowed for, a marginal preference for the coupling to the dark-energy density. Milgrom (2017) argued that the $z \approx 2$ rotation curves of Genzel et al. (2017) all but exclude an acceleration constant about four times the local one. Blanchet & Le Tiec (2009), Klinkhamer & Kopp (2011), Verlinde (2017) and Milgrom (2020) relate a_0 to Λ or to H_0 with coefficients of their own. Oppenheim & Russo (2024) argued that stochastic fluctuations of spacetime in a classical–quantum theory of gravity produce a term acting as a cosmological constant together with one of the kind used

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to fit rotation curves without dark matter; [Hertzberg & Loeb \(2024\)](#) dispute both the derivation and that the term has the MOND form. Whether one acceleration scale describes every SPARC galaxy has itself been debated: fits that marginalise over mass-to-light ratio, distance and inclination find no credible galaxy-to-galaxy variation ([Li et al. 2018](#)), whereas per-galaxy Bayesian fits were argued to exclude a common scale at more than 10σ ([Rodrigues et al. 2018a](#)), a conclusion disputed ([McGaugh et al. 2018](#); [Kroupa et al. 2018](#)), defended ([Rodrigues et al. 2018b](#)) and repeated with a different compatibility test ([Marra et al. 2020](#)). We use only the population value.

What this paper adds is not the test, which [Limbach et al. \(2008\)](#) ran, but an analysis of what the test requires: the error amplification of inferring a_0 from one galaxy, which explains why Tully–Fisher zero points and the existing high-redshift samples cannot decide; a gate in baryonic acceleration, and the halo mass it selects, which fixes the Λ CDM comparator; the number of galaxies needed, both as expected odds and as the probability of reaching them; and the result that a baryonic-mass calibration shared by the sample does not average down, with the Fisher bound that explains why deep galaxies need that calibration from outside. For κ we give three estimators with complete error budgets on a stated H_0 convention, and show that it is set by the stellar mass-to-light zero point rather than by the number of galaxies. We do not derive κ , and we take no position on whether the low-acceleration behaviour of galaxies reflects modified gravity, modified inertia or the physics of dark matter.

Every number below is produced by a public script with checks that can fail (Appendix C). We carry both densities throughout: ρ_Λ , and the critical density ρ_{crit} as the named alternative.

2 THE RELATION

The Friedmann equation for a spatially flat universe is $H^2 = 8\pi G\rho/3$. The total density today is therefore the critical density, and the part of it in the cosmological constant is

$$\rho_{\text{crit}} = \frac{3H_0^2}{8\pi G}, \quad \rho_\Lambda = \Omega_\Lambda \rho_{\text{crit}} = \frac{\Lambda c^2}{8\pi G}. \quad (2)$$

Unlike the matter density, ρ_Λ does not dilute as the Universe expands. Its expansion rate, $H_\Lambda^2 \equiv 8\pi G\rho_\Lambda/3$, gives

$$H_\Lambda = H_0\sqrt{\Omega_\Lambda} = c\sqrt{\Lambda/3}, \quad (3)$$

the constant rate that the expansion approaches in the far future.

An acceleration $a = c^\alpha G^\beta \rho^\gamma$ built from c , G and a density has, on matching the powers of kilograms, seconds and metres, the unique solution $\alpha = 1$, $\beta = \gamma = 1/2$: every such acceleration is $\kappa c\sqrt{G\rho}$ with κ a pure number. Dimensional analysis fixes the form of equation (1) and says nothing about κ ; the physical content lies in the value of κ and in the choice of ρ_Λ over another density. With equations (2) and (3) the relation can be rewritten as $a_0 = \kappa c^2\sqrt{\Lambda/8\pi} = \kappa\sqrt{3/8\pi}cH_\Lambda = \kappa M_\Lambda^2/M_{\text{P}}$ (the last with $\hbar = c = 1$, $\rho_\Lambda = M_\Lambda^4$ and $G = M_{\text{P}}^{-2}$). These are algebra; agreement among them confirms nothing, and only a measurement of a_0 does.

Table 1 evaluates equation (1) one step at a time from the [Planck Collaboration et al. \(2020\)](#) parameters, rounded to $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_\Lambda = 0.685$. For $\kappa = 1/2$,

$$a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda} = 9.36 \times 10^{-11} \text{ m s}^{-2}. \quad (4)$$

The alternative built on the critical density is larger by $\Omega_\Lambda^{-1/2} = 1.208$:

$$\frac{1}{2}c\sqrt{G\rho_{\text{crit}}} = 1.13 \times 10^{-10} \text{ m s}^{-2}. \quad (5)$$

Table 1. Equation (1) evaluated step by step. Inputs: $c = 2.99792458 \times 10^8 \text{ m s}^{-1}$, $G = 6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, $1 \text{ Mpc} = 3.08568 \times 10^{22} \text{ m}$, $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_\Lambda = 0.685$.

step	quantity	value
1	$H_0 = 67.4 \times 10^3 / 3.08568 \times 10^{22}$	$2.184 \times 10^{-18} \text{ s}^{-1}$
2	$\rho_{\text{crit}} = 3H_0^2/8\pi G$	$8.533 \times 10^{-27} \text{ kg m}^{-3}$
3	$\rho_\Lambda = \Omega_\Lambda \rho_{\text{crit}}$	$5.845 \times 10^{-27} \text{ kg m}^{-3}$
4	$G\rho_\Lambda$	$3.901 \times 10^{-37} \text{ s}^{-2}$
5	$\sqrt{G\rho_\Lambda}$	$6.246 \times 10^{-19} \text{ s}^{-1}$
6	$c\sqrt{G\rho_\Lambda}$	$1.872 \times 10^{-10} \text{ m s}^{-2}$
7	$a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$	$9.36 \times 10^{-11} \text{ m s}^{-2}$
	cH_0	$6.55 \times 10^{-10} \text{ m s}^{-2}$
	$cH_\Lambda = cH_0\sqrt{\Omega_\Lambda}$	$5.42 \times 10^{-10} \text{ m s}^{-2}$
	$\Lambda = 3H_\Lambda^2/c^2$	$1.09 \times 10^{-52} \text{ m}^{-2}$
	$R_{\text{dS}} = c/H_\Lambda$	$1.66 \times 10^{26} \text{ m} = 5.37 \text{ Gpc}$
	$M_\Lambda = \rho_\Lambda^{1/4}$	2.24 MeV

Both lie within about one systematic standard deviation of the standard determination, $a_0 = (1.20 \pm 0.02 \pm 0.24) \times 10^{-10} \text{ m s}^{-2}$ ([McGaugh et al. 2016](#)). The 21 per cent between them is the quantity that local data cannot resolve (Section 3) and that the redshift test of Section 4 magnifies to a factor of 3.8.

2.1 The classical coincidences as values of κ

From equation (3), an acceleration $a_0 = cH_\Lambda/X$ has $\kappa = \sqrt{8\pi/3}/X$, and $cH_0 = cH_\Lambda/\sqrt{\Omega_\Lambda}$. Thus the two forms of [Milgrom \(2020\)](#), $a_0 = cH_\Lambda/2\pi$ and $a_0 = cH_0/2\pi$, correspond to $\kappa = 0.461$ and $\kappa = 0.557$; the acceleration whose Unruh temperature equals the temperature of the de Sitter horizon, $a_0 = cH_\Lambda$, to $\kappa = 2.89$; the $cH_0/6$ of [Verlinde \(2017\)](#) to $\kappa = 0.583$; and the $2cH_\Lambda$ of [Milgrom \(1999\)](#) to $\kappa = 5.79$, a factor 12 too large. These are collected with the measurements in Table 6.

3 MEASURING THE COEFFICIENT

3.1 Data, fitting function and the H_0 convention

We use the SPARC database ([Lelli et al. 2016](#)): 175 disc galaxies with $3.6 \mu\text{m}$ photometry and H I/H α rotation curves. At each radius R the database gives the observed velocity V_{obs} and the Newtonian velocities V_{gas} , V_{disc} , V_{bul} of the gas and of the stellar components for unit stellar mass-to-light ratio. The two accelerations are

$$g_{\text{obs}} = \frac{V_{\text{obs}}^2}{R}, \quad g_{\text{bar}} = \frac{V_{\text{gas}}|V_{\text{gas}}| + Y_{\text{disc}}V_{\text{disc}}^2 + Y_{\text{bul}}V_{\text{bul}}^2}{R}, \quad (6)$$

where Y_{disc} and Y_{bul} are the stellar mass-to-light ratios at $3.6 \mu\text{m}$ in solar units. Keeping points with a velocity error below 10 per cent leaves 2803 points in 164 galaxies, spanning $8 \times 10^{-13} < g_{\text{bar}} < 7 \times 10^{-9} \text{ m s}^{-2}$ (Fig. 1).

The H_0 convention. The distances of 97 SPARC galaxies come from the Hubble flow with $H_0 = 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ([Lelli et al. 2016](#)), while ρ_Λ in equation (1) is built from 67.4. A distance factor f leaves g_{bar} unchanged (both V_{bar}^2 and R scale as the distance) and divides g_{obs} by f . Throughout we multiply the Hubble-flow distances by $73/67.4$, so that numerator and denominator of κ share one H_0 ; we call this the Planck-consistent convention. Table 7 gives the headline values also on SPARC’s tabulated distances and with ρ_Λ rebuilt at $H_0 = 73$. Because κ depends on H_0 through both the distances and ρ_Λ , a value of κ is incomplete without its convention (Section 3.10).

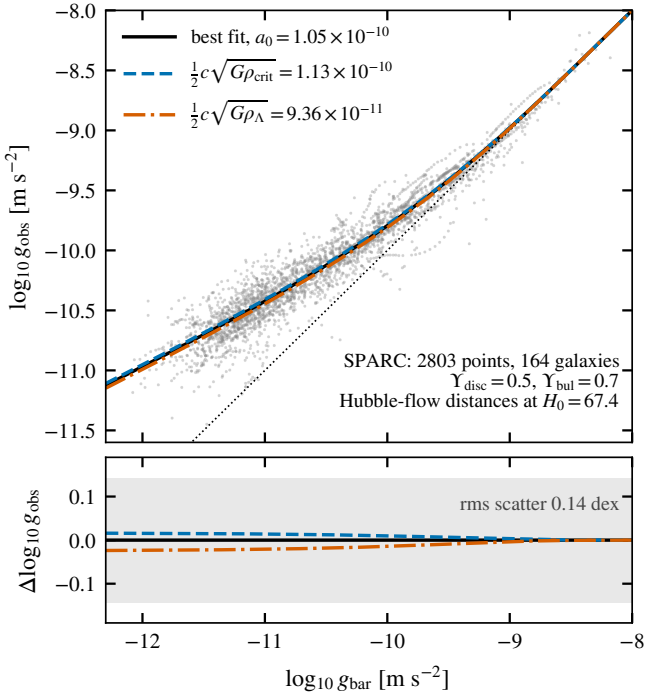


Figure 1. The radial acceleration relation of SPARC (grey points; $Y_{\text{disc}} = 0.5$, $Y_{\text{bul}} = 0.7$, Hubble-flow distances on $H_0 = 67.4$) with equation (7) for three values of a_0 : the best fit at this mass-to-light ratio (solid), $\frac{1}{2}c\sqrt{G\rho_{\text{crit}}}$ (dashed) and $\frac{1}{2}c\sqrt{G\rho_{\Lambda}}$ (dot-dashed). The dotted line is $g_{\text{obs}} = g_{\text{bar}}$. Lower panel: the curves relative to the best fit; the band is the rms scatter of the data. The curves differ by less than the scatter everywhere.

We describe the relation with the function used by [McGaugh et al. \(2016\)](#), whose form is due to [Milgrom & Sanders \(2008\)](#):

$$g_{\text{obs}} = \nu(y) g_{\text{bar}}, \quad \nu(y) = \frac{1}{1 - e^{-\sqrt{y}}}, \quad y \equiv \frac{g_{\text{bar}}}{a_0}. \quad (7)$$

For $y \gg 1$ the exponential vanishes, $\nu \rightarrow 1$ and $g_{\text{obs}} = g_{\text{bar}}$. For $y \ll 1$, put $s = \sqrt{y}$ and expand $1 - e^{-s} = s - s^2/2 + \dots$; then

$$\nu = \frac{1}{s} \left(1 + \frac{s}{2} + \dots\right) \implies g_{\text{obs}} = \sqrt{g_{\text{bar}} a_0} \left(1 + \frac{1}{2}\sqrt{y} + \dots\right). \quad (8)$$

The leading term is the deep regime, in which $a_0 = g_{\text{obs}}^2/g_{\text{bar}}$. The correction falls only as $\sqrt{y}$: at $y = 0.01$ it is still 5 per cent in g_{obs} and 10 per cent in a_0 . This slow approach matters repeatedly below. We use equation (7) as an empirical description of galaxy data and nothing more; Section 5 returns to this.

3.2 The standard fit and its degeneracy with the mass-to-light ratio

The simplest estimate minimises the residuals in $\log_{10} g_{\text{obs}}$ over a_0 at fixed mass-to-light ratios. On SPARC’s tabulated distances with $Y_{\text{disc}} = 0.5$ and $Y_{\text{bul}} = 0.7$ we recover the value of [McGaugh et al. \(2016\)](#), $a_0 = 1.19 \times 10^{-10} \text{ m s}^{-2}$ with an rms scatter of 0.14 dex. On the Planck-consistent distances the same fit gives $1.05 \times 10^{-10} \text{ m s}^{-2}$, 12 per cent lower. The result depends steeply on Y_{disc} (Table 2, Fig. 2a), and equation (8) shows why. Where stars dominate, $g_{\text{bar}} \propto Y$; in the deep regime the data fix the product $g_{\text{bar}} a_0 = g_{\text{obs}}^2$; so $a_0 \propto 1/Y$.

The quality of the fit is almost flat across the table, so the data do

Table 2. The standard fit of equation (7) to SPARC as a function of the disc mass-to-light ratio ($Y_{\text{bul}} = 0.7$; Planck-consistent distances). κ_{Λ} and κ_{crit} are a_0 divided by $c\sqrt{G\rho_{\Lambda}}$ and by $c\sqrt{G\rho_{\text{crit}}}$.

Y_{disc}	$a_0 [10^{-10} \text{ m s}^{-2}]$	κ_{Λ}	κ_{crit}	rms [dex]
0.40	1.32	0.703	0.582	0.147
0.50	1.05	0.560	0.464	0.143
0.60	0.86	0.457	0.379	0.143
0.70	0.71	0.381	0.315	0.145
0.80	0.60	0.322	0.266	0.150

not choose Y_{disc} . Stellar population models give $Y_{\text{disc}} \approx 0.5$ – 0.7 at $3.6 \mu\text{m}$ ([Meidt et al. 2014](#); [Schombert et al. 2019](#)). In this estimator $\kappa = 1/2$ requires $Y_{\text{disc}} = 0.555$ for the ρ_{Λ} reading and 0.465 for the ρ_{crit} reading, the latter just below the population range (on the tabulated distances, 0.621 and 0.524). That is all the standard fit can say. Restricting it to the deep band, $g_{\text{bar}} < 0.1a_0$, where the stars carry less of the mass, does not remove the dependence: $\kappa = 0.50$, 0.44 and 0.40 for $Y_{\text{disc}} = 0.5$, 0.6 and 0.7. (Giving the points whose velocity errors exceed 10 per cent equal weight lowers each value by about 0.1.) Three estimators that depend on the mass-to-light ratio in different ways do better.

3.3 Estimator A: the Tully–Fisher intercept, with its finite-depth correction

The baryonic Tully–Fisher relation is often quoted as $V_f^4 = GM_b a_0$ for the flat rotation velocity V_f and the baryonic mass M_b . That is a limit. The exact statement follows from equation (7) at the outermost measured radius R . With $g_{\text{obs}} = V_f^2/R = \nu(y)g_{\text{bar}}$, write $g_{\text{bar}}^2 R^2 = y a_0 \times \Gamma GM_b$, where the geometry factor $\Gamma \equiv g_{\text{bar}} R^2/(GM_b)$ equals one for a point mass, exceeds one for a disc at finite radius and is read from the SPARC mass model (median 1.19). Then

$$V_f^4 = Q(y) \Gamma GM_b a_0, \quad Q(y) \equiv \nu^2(y) y = 1 + \sqrt{y} + \dots, \quad (9)$$

so that

$$\hat{a}_0 = \frac{V_f^4}{Q(y) \Gamma GM_b}. \quad (10)$$

The uncorrected ratio V_f^4/GM_b overestimates a_0 by the factor $Q\Gamma$. On SPARC this factor is not small. Of the 85 galaxies with flat outer rotation curves, the 24 deepest have outermost accelerations $y < 0.05$; for them, on SPARC’s tabulated distances, the uncorrected ratio is $1.15 \times 10^{-10} \text{ m s}^{-2}$, the typical corrections are $\Gamma \approx 1.19$ and $Q \approx 1.18$, and the corrected, weighted value is $\hat{a}_0 = 8.7 \times 10^{-11} \text{ m s}^{-2}$ ($\kappa = 0.465$). The estimate depends on the Hubble-flow distances through their weight in the sample, 0.20 (our repository’s convention audit), so $\hat{a}_0 \propto D_{\text{HF}}^{-0.40}$, and on the Planck-consistent convention

$$\kappa_{\Lambda} = 0.450 \pm 0.073. \quad (11)$$

The error budget (Table 3) is not dominated by statistics. Its largest term is the form of the transition function. Because of the slow approach in equation (8), functions that reach the deep regime faster than equation (7) have Q closer to one and return larger values, up to $\kappa = 0.55$. The deepest SPARC galaxy has $y = 0.012$; the function would stop mattering only below $y \approx 0.003$.

3.4 A floor that no sample removes

Two of the correlated terms cannot be reduced by choosing the sample. The baryonic mass is $M_b = YL + M_{\text{gas}}$. Let f_* be the stellar

Table 3. Error budget of estimator A (24 galaxies; effective number 10.1, because five galaxies carry 61 per cent of the weight). Entries are fractional errors on a_0 in per cent. The distance-scale term already exceeds the 3.2 per cent shift between SPARC’s tabulated distances and the Planck-consistent convention, so the shift is applied to the central value and not added to the error.

<i>random terms</i> (decrease with sample size)	
distances, galaxy by galaxy	4.8
flat velocity	4.1
inclination	3.9
scatter in Υ between galaxies	3.1
H I fluxes	1.2
<i>quadrature sum</i> (bootstrap: 8.6)	8.1
<i>correlated terms</i> (independent of sample size)	
form of the transition function	8.1
zero point of Υ at 3.6 μm (0.5–0.7)	6.9
H I flux scale (10 per cent)	5.9
distance scale (H_0 and TRGB zero point)	5.1
CO-dark molecular gas (0–10 per cent)	3.0
definition of the flat velocity	2.2
helium correction (1.33–1.40)	1.5
<i>quadrature sum</i>	13.8
total	16.2

fraction of M_b , s_Υ the fractional uncertainty of the stellar mass scale and s_g that of the gas mass scale. The fractional error on M_b , and so on a_0 in equation (10), is

$$\sigma(f_*) = \sqrt{(f_* s_\Upsilon)^2 + [(1 - f_*) s_g]^2}. \quad (12)$$

Choosing gas-rich galaxies lowers f_* and suppresses the first term, but it hands the budget to the second. The minimum follows from $d\sigma^2/df_* = 0$:

$$f_* = \frac{s_g^2}{s_\Upsilon^2 + s_g^2}, \quad \sigma_{\min} = \frac{s_\Upsilon s_g}{\sqrt{s_\Upsilon^2 + s_g^2}}. \quad (13)$$

With $s_\Upsilon = 0.17$ (the range 0.5–0.7 about 0.6) and $s_g = 0.115$ (H I flux scale, helium and CO-dark gas in quadrature) the floor is $\sigma_{\min} = 9.5$ per cent, reached at $f_* = 0.32$. It does not depend on the number of galaxies, on the transition function, or on distances. Even if both scales were known to 5 per cent the floor would be 3.5 per cent. The route to a better κ therefore runs through the calibration of stellar and gas masses, not through larger samples.

3.5 Estimator B: the shape of the relation, immune to a common distance scale

Suppose all distances are wrong by a common factor. In the plane of Fig. 1 that is a purely vertical shift, because g_{bar} does not change and $g_{\text{obs}} \propto 1/D$. We therefore fit

$$\log_{10} g_{\text{obs}} = \log_{10} [g_{\text{bar}} \nu(g_{\text{bar}}/a_0)] + C \quad (14)$$

with the offset C free. The offset absorbs the distance scale, and a_0 is fixed by where the bend lies along the g_{bar} axis. On SPARC, rescaling every distance by 5 or 10 per cent changes this a_0 by less than one part in 10^{10} , while the standard fit moves by 22 per cent for a 10 per cent rescaling.

The price of the immunity is a strong dependence on the mass-to-light ratios, above all that of the bulges, because the bend is located using the high-acceleration points where bulges dominate. Table 4 gives the result over the ranges $\Upsilon_{\text{disc}} = 0.5\text{--}0.7$ and $\Upsilon_{\text{bul}} = 0.6\text{--}0.8$. At the centre of the grid $\kappa = 0.584$. The half-range over the grid

Table 4. Estimator B (shape-only; Planck-consistent distances): κ as a function of the two stellar mass-to-light ratios.

	$\Upsilon_{\text{bul}} = 0.6$	0.7	0.8
$\Upsilon_{\text{disc}} = 0.5$	0.444	0.567	0.715
$\Upsilon_{\text{disc}} = 0.6$	0.470	0.584	0.718
$\Upsilon_{\text{disc}} = 0.7$	0.502	0.610	0.735

is 0.146, of which the bulge term alone is 0.124 and the disc term 0.022. An 11.5 per cent change of the gas scale moves κ by 0.048, and a bootstrap over galaxies gives a statistical error of 0.102, far larger than the formal error of the fit, because a few bulge-dominated galaxies carry the high-acceleration end. In quadrature,

$$\kappa_B = 0.58 \pm 0.18. \quad (15)$$

An earlier version of this estimator held Υ_{bul} fixed and used the formal statistical error and SPARC’s tabulated distances, which gave 0.551 ± 0.043 ; that error bar was too small by a factor of four. Restricting the fit to the 132 galaxies without bulges does not help. Their baryonic accelerations stop at $16a_0$, the lever arm is too short to locate the bend, and κ runs from 0.51 to 1.13 across $\Upsilon_{\text{disc}} = 0.5\text{--}0.7$. The immunity is also to a *common* rescaling only. The Hubble-flow galaxies lie preferentially at the high-acceleration end of the relation, where the estimator locates the bend, so the H_0 convention moves it: from 0.584 (Planck-consistent) to 0.547 on the tabulated distances and to 0.489–0.505 with ρ_Λ rebuilt at $H_0 = 73$ (Table 7). Estimator B is therefore a cross-check on the distance scale and a poor measurement of κ .

3.6 Estimator C: a profile likelihood with the mass-to-light ratio free for every galaxy

The degeneracy between Υ and a_0 is exact in the deep regime only if every baryonic component scales with Υ . The gas does not, and the transition region does not, so freeing the stellar mass-to-light ratio galaxy by galaxy still leaves a_0 constrained. We profile the likelihood of equation (7) over all SPARC points ($V > 0$, velocity errors clipped at 1 km s^{-1}), with Υ_{disc} free for each galaxy, $\Upsilon_{\text{bul}} = 1.4 \Upsilon_{\text{disc}}$, the gas fixed, and an intrinsic scatter (0.077 dex) set by $\chi^2/\text{d.o.f.} = 1$ at the best fit. Unlike the ratios of the pitfall in Section 3.8, these are fitted jointly with a_0 . On the Planck-consistent distances the best fit for 175 galaxies (3391 points) is $a_0 = 8.1 \times 10^{-11} \text{ m s}^{-2}$, with a galaxy bootstrap error of 9.3 per cent. Two systematic terms dominate. An 11.5 per cent change of the gas scale, to which an estimate anchored by the gas is directly exposed, moves κ by 0.039. And the transition function matters: the same likelihood with $g_{\text{obs}}^2 = g_{\text{bar}}^2 + g_{\text{bar}} a_0$ gives $\kappa = 0.54$, so half the difference, 0.053, is carried as a term. In quadrature,

$$\kappa_C = 0.43 \pm 0.08. \quad (16)$$

An earlier analysis in our repository used the second function and SPARC’s tabulated distances and found $a_0 = 1.08 \times 10^{-10} \text{ m s}^{-2}$ ($\kappa_\Lambda = 0.575$); our code reproduces that result exactly when given its inputs, and the difference from equation (16) is the two choices just named.

3.7 Do the estimators measure, or echo an assumed value?

Every estimator above contains a_0 somewhere in its machinery, so we test them on synthetic data. We keep the real SPARC baryons, generate g_{obs} from equation (7) with a known a_0 equal to 0.7, 1.0

and 1.3 times the value of equation (4), and add the quoted velocity errors plus 0.034 dex of intrinsic scatter. The standard fit returns the injected value to within 1 per cent and the deep-band fit to within 2.1 per cent. The shape-only fit returns it to within 2.3 per cent even when every g_{obs} is shifted by a common 0.08 dex, the effect of a 20 per cent distance error, which moves the standard fit by 60 to 66 per cent. Estimator A evaluates its correction $Q(y)$ at an assumed a_0 ; this is a sensitivity test on the real data rather than an injection. Iterating the correction to the estimator's own output changes the result by 0.6 per cent, and raising the assumed value by 21 per cent changes it by 9 per cent. None of the measured values is an echo of an assumed one.

3.8 The deep regime in two surveys: slope and amplitude

Below a_0 equation (7) predicts two things: how steeply g_{obs} rises with g_{bar} , and where the curve lies. With $n(y) \equiv d \ln v / d \ln y$,

$$\beta(y) \equiv \frac{d \ln g_{\text{obs}}}{d \ln g_{\text{bar}}} = 1 + n(y) = 1 - \frac{\sqrt{y}/2}{e^{\sqrt{y}} - 1} = \frac{1}{2} + \frac{\sqrt{y}}{4} + \dots \quad (17)$$

The slope tends to one half and approaches it just as slowly: $\beta = 0.525, 0.554, 0.575$ and 0.603 at $y = 0.01, 0.05, 0.1$ and 0.2 . A measured deep slope must therefore be compared with equation (17) at the accelerations actually sampled, not with one half. Moving a_0 from the ρ_Λ value to the ρ_{crit} value changes the expected slope at the SPARC points by less than 0.01, so the slope tests the form of the relation and the amplitude tests κ .

We use the window $g_{\text{bar}} < 0.2a_0 = 1.87 \times 10^{-11} \text{ m s}^{-2}$, with a_0 fixed at the value of equation (4), in two surveys with independent data and mass models. The first is SPARC, as above. The second is MIGHTEE-HI (Vărășteanu et al. 2025): 19 H I-selected galaxies at $z < 0.08$ with MeerKAT rotation curves and stellar masses from resolved fits to their spectral energy distributions (SED). The survey publishes its radial acceleration relation only as a figure. We digitised its 80 points from the vector graphics; fitting equation (7) to them returns the survey's own value of a_0 to 1.2 per cent. We identify galaxies by marker colour, which is not a clean identifier: the figure has 18 colours for 19 galaxies, and the largest group, 11 points spanning only 0.22 dex in g_{bar} , may hold two. The window holds 72 of the points, 70 of them in the 17 groups with at least three.

The slope. For each galaxy with at least three points in the window we fit a straight line to $\log g_{\text{obs}}$ against $\log g_{\text{bar}}$, and we take the median over galaxies. Offsets that are constant within a galaxy, from its distance, its inclination or its stellar mass-to-light ratio, do not bias this statistic, so it does not depend on the H_0 convention. The expectation is the same statistic computed from $g_{\text{bar}} v(g_{\text{bar}}/a_0)$ at the same points, and a bootstrap over galaxies gives the error of the difference. Table 5 and Fig. 3(a) show the result. SPARC gives $\beta = 0.58$ – 0.62 against an expectation of 0.57, and MIGHTEE-HI gives 0.60 ± 0.08 against 0.56 (0.60 ± 0.09 against 0.56 without the largest colour group); the differences lie between 0.1 and 1.6σ . Together the two surveys exclude a slope of 0.75 at 4.2 – 4.9σ and the Newtonian slope at 12σ . On synthetic data built on the real baryons the statistic is unbiased to 0.005, and it returns an injected slope of 0.75 as 0.751 (SPARC) and 0.752 (MIGHTEE-HI). A single straight line through all the SPARC points in the window, which is exposed to the galaxy-to-galaxy offsets, also agrees: 0.52 – 0.57 against 0.56. The survey's own fit finds a low-acceleration slope near one half (Vărășteanu et al. 2025).

The amplitude. Fitting equation (7) in the window gives, on SPARC, $\kappa_\Lambda = 0.54 \pm 0.04, 0.47 \pm 0.03$ and 0.41 ± 0.03 for $Y_{\text{disc}} = 0.5, 0.6$ and 0.7 (bootstrap over galaxies), a range that contains one half.

Table 5. The deep regime, $g_{\text{bar}} < 0.2a_0$. β is the median over galaxies of the slope of $\log g_{\text{obs}}$ against $\log g_{\text{bar}}$; β_v is the same statistic computed from equation (7) at the same points, with $\kappa_\Lambda = 1/2$; Δ/σ is their difference in units of its bootstrap error. κ_Λ is equation (7) fitted in the window (SPARC on the Planck-consistent distances), except in the rows marked V25, which are the fits of Vărășteanu et al. (2025) to all radii under their stated mass-to-light conventions. The last row uses mass-to-light ratios fitted to the rotation curves themselves (Section 3.8).

sample	N_{gal}	β	β_v	Δ/σ	κ_Λ
SPARC, $Y_{\text{disc}} = 0.5$	125	0.62 ± 0.03	0.570	+1.6	0.54 ± 0.04
SPARC, $Y_{\text{disc}} = 0.6$	120	0.61 ± 0.03	0.573	+1.3	0.47 ± 0.03
SPARC, $Y_{\text{disc}} = 0.7$	116	0.58 ± 0.04	0.575	+0.1	0.41 ± 0.03
MIGHTEE-HI, digitised	17	0.60 ± 0.08	0.556	+0.5	0.89 ± 0.13
V25, SED ratios	19				0.90 ± 0.07
V25, no H_2	19				1.10 ± 0.08
V25, radial mean	19				0.79 ± 0.07
V25, $Y_K = 0.6$	19				0.58 ± 0.05
SPARC, fitted ratios					0.25 ± 0.03

MIGHTEE-HI gives $\kappa_\Lambda = 0.89 \pm 0.13$, and the survey's own fit to all radii gives 0.90 ± 0.07 . Taken at face value, the survey's own value and SPARC disagree by 5.0 to 7.5σ in $\ln a_0$. The disagreement lies mostly in the stellar mass-to-light ratio. The SED fits of MIGHTEE-HI have a median of 0.35 in the K_s band, whereas another recent determination in the same band finds a median of 0.72 (Marasco et al. 2025). When the survey fixes $Y_K = 0.6$ it obtains $a_0 = (1.08 \pm 0.09) \times 10^{-10} \text{ m s}^{-2}$, that is $\kappa_\Lambda = 0.58 \pm 0.05$, which lies 0.7 and 2.0σ above SPARC at $Y_{\text{disc}} = 0.5$ and 0.6 but 3.1σ above it at 0.7 (Fig. 3b; the comparison also carries any difference between the two surveys' distance conventions). The deep regime therefore agrees with $\kappa \approx 1/2$ when the two surveys use comparable mass-to-light ratios, its amplitude is set by that choice rather than by the relation, and it cannot separate the ρ_Λ reading from the ρ_{crit} reading.

A pitfall. The amplitude is only as good as the stellar masses. Mass-to-light ratios fitted galaxy by galaxy to the rotation curves without the relation absorb part of the mass discrepancy into the stars: in one such compilation of SPARC disc ratios (median 1.17, up to 15) their logarithms correlate at $r = 0.92$ with those of the ratio at which the baryons alone reproduce each curve. Used in the window fit, they give $\kappa_\Lambda = 0.25 \pm 0.03$, a factor of 1.7 to 2.2 below the population ratios, and they manufacture an apparent deficit in the deep regime.

3.9 What the measurements say

Table 6 and Fig. 2(b) compare the three estimators with the candidate coefficients, and Table 7 gives each on four H_0 conventions. On the Planck-consistent convention $\kappa = 1/2$ lies 0.7σ from estimator A, 0.5σ from B and 0.9σ from C. The coincidence $cH_\Lambda/2\pi$ does as well, and on a common footing Milgrom's 2π forms are at least as close to A and C as $1/2$ is: $cH_\Lambda/2\pi$ against $1/2$ on the ρ_Λ footing, and $cH_0/2\pi$ (1.5 and 1.6σ) against $\frac{1}{2}c\sqrt{G\rho_{\text{crit}}}$ (2.1 and 2.2σ) on the critical density. The estimators share galaxies and calibrations, so we neither average them nor multiply their likelihoods. Taken one at a time, the likelihood ratio between $\kappa = 1/2$ and 0.461 is $e^{-0.22}$ (A), $e^{+0.12}$ (B) and $e^{-0.31}$ (C): the data do not distinguish them. The thermal identifications are excluded: $a_0 = cH_\Lambda$ by a factor of six and the $2cH_\Lambda$ of Milgrom (1999) by a factor of 12.

Published determinations, converted in Table 8, show that the inferred coefficient follows the treatment of the stellar mass-to-light ratio. Against our working value: the full joint inference of Desmond

Table 6. Candidate coefficients in $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ and their distance, in standard deviations, from estimators A (0.450 ± 0.073), B (0.58 ± 0.18) and C (0.43 ± 0.08), all on the Planck-consistent convention. The last row is $\kappa = 1/2$ on the critical density, expressed as κ_Λ .

a_0	κ	A	B	C
$cH_\Lambda/2\pi$ (Milgrom 2020)	0.461	+0.1	-0.7	+0.4
$\frac{1}{2}c\sqrt{G\rho_\Lambda}$ (working value)	0.500	+0.7	-0.5	+0.9
$cH_0/2\pi$ (Milgrom 2020)	0.557	+1.5	-0.1	+1.6
$cH_0/6$ (Verlinde 2017)	0.583	+1.8	0.0	+1.9
cH_Λ (Unruh = de Sitter temperature)	2.89	+33	+13	+32
$2cH_\Lambda$ (Milgrom 1999)	5.79	+73	+28	+70
$\frac{1}{2}c\sqrt{G\rho_{\text{crit}}}$	0.604	+2.1	+0.1	+2.2

Table 7. The three estimators of κ_Λ on four H_0 conventions. Planck-consistent (adopted): SPARC’s Hubble-flow distances moved to $H_0 = 67.4$. Tabulated: SPARC’s distances as published (Hubble flow at 73) with ρ_Λ at 67.4. R2a and R2b: SPARC’s distances with ρ_Λ rebuilt at $H_0 = 73$, holding Ω_Λ or $\Omega_m h^2$ fixed. The conventions are alternatives, not errors.

estimator	Planck-consistent	tabulated	R2a	R2b
A: Tully–Fisher intercept	0.450	0.465	0.429	0.416
B: shape only	0.584	0.547	0.505	0.489
C: profile likelihood	0.434	0.469	0.433	0.419

(2023) over distances, inclinations, luminosities and mass-to-light ratios finds $a_0 = (1.19 \pm 0.04 \pm 0.09) \times 10^{-10} \text{ m s}^{-2}$, which is $\kappa_\Lambda = 0.636 \pm 0.053$; it puts $\kappa = 1/2$ at 2.6σ on the ρ_Λ footing (and 0.461 at 3.3σ), and $\frac{1}{2}c\sqrt{G\rho_{\text{crit}}}$ at 0.6σ (and $cH_0/2\pi$ at 1.5σ) on the critical density. That inference, like McGaugh et al. (2016), starts from SPARC’s tabulated distances, on which our standard fit is 12 per cent higher than on the Planck-consistent convention; we do not convert it, because its distances are free parameters with priors. The MIGHTEE-HI determinations of Vărașteanu et al. (2025, 2026), $\kappa_\Lambda = 0.80\text{--}0.90$ with their SED-based ratios and 0.58 with $Y_K = 0.6$, make the same point in an independent survey.

We adopt $\kappa = 1/2$ as the working value because it is the simplest number consistent with our three estimators, and for no deeper reason: it is fitted, and a published joint inference disfavours it at 2.6σ on its own footing. Separating it from 0.461 at 3σ would need a_0 to 2.6 per cent, below the floor of equation (13). The deep regime of Section 3.8 is a consistency test rather than a fourth estimator, because its amplitude follows the stellar mass-to-light convention.

3.10 The coefficient is degenerate with the Hubble constant

Substituting ρ_{crit} into equation (1) gives $a_0 = \kappa c H_0 \sqrt{3\Omega_\Lambda/8\pi}$: at fixed Ω_Λ the predicted acceleration depends only on the product κH_0 . The two values of the Hubble constant currently in tension, 67.4 (Planck Collaboration et al. 2020) and $73.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Riess et al. 2022), have the ratio 0.923 , and $0.461/0.500 = 0.921$, so $(\kappa, H_0) = (0.500, 67.4)$ and $(0.461, 73.04)$ predict the same a_0 to 0.2 per cent (Fig. 2c); with $\Omega_m h^2$ held at its CMB value instead, the matching coefficient is 0.446 . The first statement is exact; the second is a numerical coincidence to which we attach no meaning. A claim that the coefficient is ‘exactly one half’ is also a claim about H_0 and about the distance scale of the galaxies, which is why Section 3.1 fixes the convention. Read in reverse at $\kappa = 1/2$, with the Hubble-flow distances rebuilt at each trial H_0 , estimator A gives $H_0 = 57 \pm 15$ and estimator B $73 \pm 15 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (their sensitivities to those distances have opposite signs), far too coarse to bear on the tension.

4 THE REDSHIFT TEST

The value of κ cannot distinguish ρ_Λ from ρ_{crit} , nor either from an emergent scale. The redshift dependence can. We define

$$\Delta(z) \equiv \log_{10} \frac{a_0(z)}{a_0(0)}. \quad (18)$$

4.1 Three laws

(a) *A scale anchored to Λ .* The cosmological constant does not evolve, so ρ_Λ and a_0 are the same at every redshift:

$$\Delta_\Lambda(z) = 0. \quad (19)$$

This is the content of the hypothesis, not evidence for it. Plain MOND, in which a_0 is simply a constant of nature, makes the same prediction. What the Λ anchor adds is a reason for the constancy, and with it the exclusion of law (b).

(b) *A scale anchored to the critical density.* Replacing ρ_Λ by $\rho_{\text{crit}}(z) = 3H^2(z)/8\pi G$ gives $a_0 \propto H(z)$. With $E(z) \equiv H(z)/H_0 = [\Omega_m(1+z)^3 + \Omega_\Lambda]^{1/2}$,

$$\Delta_H(z) = \log_{10} E(z). \quad (20)$$

At $z = 2.5$, $E^2 = 0.315 \times 3.5^3 + 0.685 = 14.19$, so $E = 3.77$ and $\Delta_H = +0.58$. At $z = 1.2$, the limit of Limbach et al. (2008), it is $+0.30$; at $z = 2$ the factor is 3.0 , below the factor of about four that Milgrom (2017) considers all but excluded by the Genzel et al. (2017) discs.

(c) *The emergent scale of Λ CDM haloes.* In Λ CDM the characteristic acceleration of the relation is inherited from dark-matter haloes. Appendix A shows that the central acceleration of an NFW halo (Navarro et al. 1997) of mass M_{200} and concentration c_{200} is

$$g_{\text{max}} = \frac{100^{2/3}}{2} (GM_{200})^{1/3} H^{4/3}(z) \frac{c_{200}^2}{f(c_{200})}, \quad (21)$$

where $f(x) = \ln(1+x) - x/(1+x)$. For $M_{200} = 10^{12} M_\odot$ today this is $8.1 \times 10^{-11} \text{ m s}^{-2}$, close to a_0 , which is part of the reason that galaxies in Λ CDM haloes show an a_0 -like scale (Navarro et al. 2017). At fixed halo mass,

$$\Delta_{\text{halo}}(z) = \frac{4}{3} \log_{10} E(z) + \log_{10} \frac{[c^2/f(c)]_z}{[c^2/f(c)]_0}. \quad (22)$$

The first term rises, because haloes of a given mass were denser in the past. The second falls, because they were less concentrated. With the concentration–mass relation of Dutton & Macciò (2014), c_{200} of a $10^{12} M_\odot$ halo falls from 8.4 to 3.9 between $z = 0$ and 2.5 , and $\Delta_{\text{halo}}(2.5) = 0.768 - 0.440 = +0.33$, a factor 2.13 . That number should not be used as the decision value, because it depends on the halo mass: over $M_{200} = 10^{11}\text{--}10^{13} M_\odot$ and the relations of Dutton & Macciò (2014) and Duffy et al. (2008) it runs from $+0.23$ to $+0.46$ (Table 9). The halo mass that matters is the one the measurement selects. The gate of Section 4.4 admits discs with $V_f \approx 83\text{--}125 \text{ km s}^{-1}$; at $z = 2.5$, for $V_f/V_{200} = 1.0\text{--}1.2$, their haloes have $M_{200} = 3 \times 10^{10}\text{--}1.8 \times 10^{11} M_\odot$, where the Dutton–Macciò relation gives $+0.17$ to $+0.25$ dex. We adopt the value for the central example ($M_{200} = 7.9 \times 10^{10} M_\odot$),

$$\Delta_{\text{halo}}(2.5) = +0.22, \quad (23)$$

as the decision value. The choice of concentration–mass relation is a model systematic of the same size: Duffy et al. (2008) give $+0.43$ for the same haloes. A different structural scaling, the Tully–Fisher ratio V_{max}^4/GM at a fixed ratio of baryonic to halo mass, gives $+0.56$ for

Table 8. Determinations of a_0 converted to κ on the two footings, with each analysis’s treatment of the stellar mass-to-light ratio. Published values are converted as published (their own distance conventions); errors combine statistical and systematic terms in quadrature where both are given. This paper’s values are on the Planck-consistent convention.

analysis	sample	mass-to-light ratio	a_0 [10^{-10} m s $^{-2}$]	κ_Λ	κ_{crit}
McGaugh et al. (2016)	SPARC	fixed: 0.5 / 0.7	1.20 ± 0.24	0.64 ± 0.13	0.53 ± 0.11
Desmond (2023)	SPARC	free (joint inference)	1.19 ± 0.10	0.64 ± 0.05	0.53 ± 0.04
Vărașteanu et al. (2025)	MIGHTEE-HI	resolved SED fits	1.69 ± 0.13	0.90 ± 0.07	0.75 ± 0.06
Vărașteanu et al. (2025)	MIGHTEE-HI	$\Upsilon_K = 0.6$ fixed	1.08 ± 0.09	0.58 ± 0.05	0.48 ± 0.04
Vărașteanu et al. (2026)	MIGHTEE-HI/LADUMA	fiducial (varying)	1.50 ± 0.05	0.80 ± 0.03	0.66 ± 0.02
this paper, estimator A	SPARC	0.5–0.7 (budget)	0.84 ± 0.14	0.45 ± 0.07	0.37 ± 0.06
this paper, estimator B	SPARC	grid, Table 4	1.09 ± 0.35	0.58 ± 0.18	0.48 ± 0.15
this paper, estimator C	SPARC	free per galaxy	0.81 ± 0.14	0.43 ± 0.08	0.36 ± 0.06

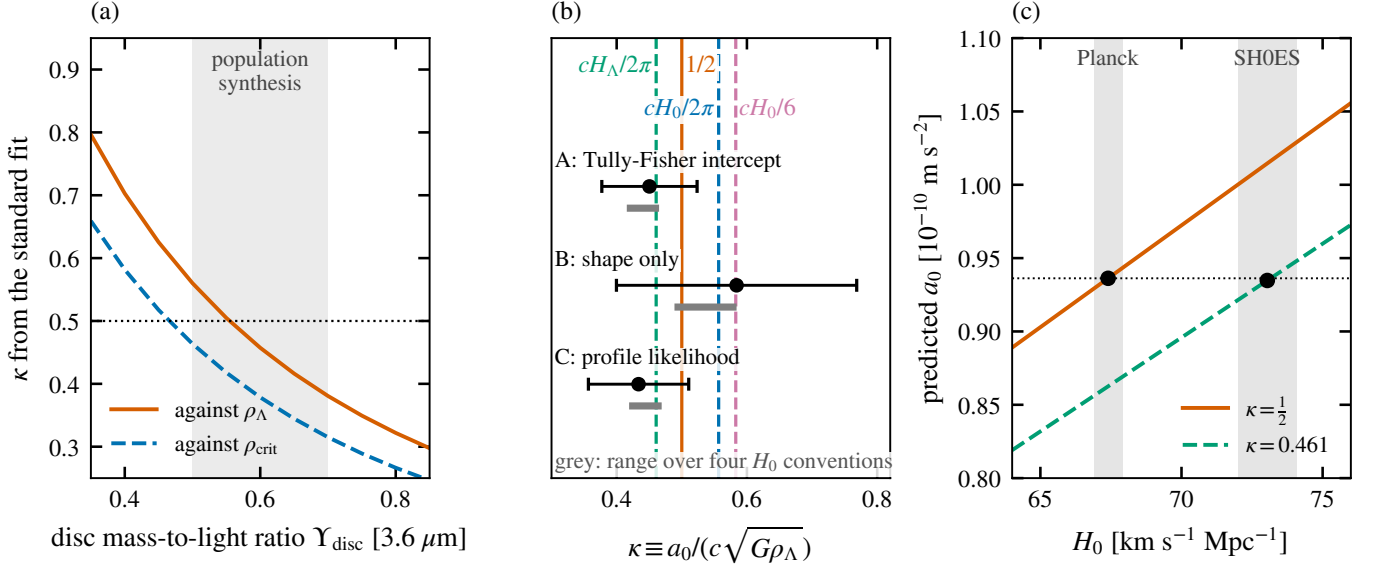


Figure 2. (a) The coefficient returned by the standard fit as a function of the disc mass-to-light ratio, measured against ρ_Λ (solid) and against ρ_{crit} (dashed), on the Planck-consistent distances; the band is the range of stellar population models. (b) Estimators A, B and C (points with 1σ total errors, Planck-consistent convention) against the candidate coefficients of Table 6. Grey bars: the full range of each estimator over the four H_0 conventions of Table 7, computed by the same scripts; they are alternatives rather than errors, so they need not be symmetric about the point. (c) The H_0 lock: the a_0 predicted by $\kappa = 1/2$ (solid) and by $\kappa = 0.461$ (dashed) as a function of H_0 at fixed Ω_Λ . The two black points, at the Planck and SH0ES values of H_0 , coincide to 0.2 per cent.

$10^{12} \text{ M}_\odot$ (Appendix A). Hydrodynamical simulations point the same way: Keller & Wadsley (2017) find that the simulated relation is not constant with redshift, and in the Magneticum simulations the best-fitting a_0 rises by a factor of about three between $z = 0$ and 2 (Mayer et al. 2023), which is $+0.48$ dex. Every Λ CDM estimate is a rise of 0.17 – 0.6 dex at $z = 2.5$. The g_{max} proxy is itself a heuristic, not a Λ CDM prediction for the statistic of the decision rule; calibrating the Λ CDM side on simulated low-mass discs inverted in the same way at $y < 0.3$ would replace it and is not done here. Section 4.4 carries the relation spread as a systematic of the design.

(d) *If dark energy evolves.* Law (a) assumes a true cosmological constant. If instead a_0 followed an evolving dark-energy density, $a_0 \propto \sqrt{\rho_{\text{DE}}(z)}$, then for the $w_0 w_a$ fits of DESI Collaboration et al. (2025)

$$\frac{\rho_{\text{DE}}(z)}{\rho_{\text{DE}}(0)} = (1+z)^{3(1+w_0+w_a)} \exp\left[-\frac{3w_a z}{1+z}\right] \quad (24)$$

gives $\Delta = -0.09$ to -0.11 at $z = 2.5$ for the three supernova combinations. This is not our prediction. We quote it to show that the test is robust to the question: the alternatives to constancy differ from it by $+0.2$ to $+0.6$ dex, while this differs by -0.1 .

Table 9. $\Delta(z) = \log_{10}[a_0(z)/a_0(0)]$ in dex for the laws of Section 4.1. Column 4: the halo law for $M_{200} = 10^{12} \text{ M}_\odot$ with the concentrations of Dutton & Macciò (2014) (used for the massive discs of Section 4.3); column 5: the range over 10^{11} – $10^{13} \text{ M}_\odot$ and the relations of Dutton & Macciò (2014) and Duffy et al. (2008); column 6: the halo law for the gate’s central disc ($V_f = 105 \text{ km s}^{-1}$, $V_f/V_{200} = 1.1$), whose halo mass is set at each redshift. The last column is the density mapping of Section 4.1(d) for the DESI+CMB+DESY5 fit.

z	Λ	$H(z)$	halo	halo, range	halo, gate	$\sqrt{\rho_{\text{DE}}}$
1.0	0	+0.25	+0.09	[+0.05, +0.16]	+0.06	+0.00
2.0	0	+0.48	+0.25	[+0.16, +0.37]	+0.16	−0.07
2.5	0	+0.58	+0.33	[+0.23, +0.46]	+0.22	−0.10
3.0	0	+0.66	+0.41	[+0.29, +0.54]	+0.27	−0.13

4.2 The statistic, and why the depth of the galaxy matters

Given g_{obs} and g_{bar} at one radius of one galaxy, equation (7) can be solved for a_0 exactly: find the y for which $v(y) = g_{\text{obs}}/g_{\text{bar}}$ and set

$$\hat{a}_0 = \frac{g_{\text{bar}}}{y}. \quad (25)$$

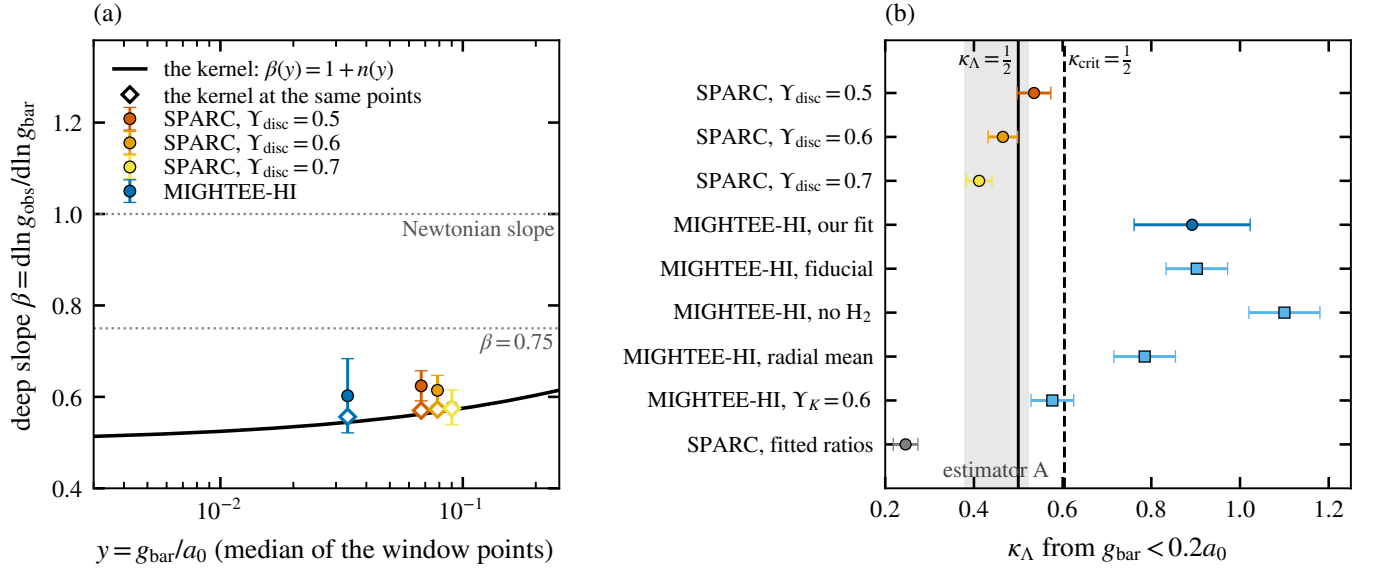


Figure 3. The deep regime in two surveys. (a) The slope of the relation. The curve is equation (17). Filled points are the median per-galaxy slopes of SPARC, at three disc mass-to-light ratios, and of MIGHTEE-HI in the window $g_{\text{bar}} < 0.2 a_0$, placed at the median y of their points (the SPARC points are offset by ± 10 per cent for clarity), with bootstrap errors. Open diamonds are the same statistic computed from equation (7) at the same points. The dotted lines mark a slope of 0.75 and the Newtonian slope. (b) The coefficient in the deep regime: SPARC at three disc ratios (Planck-consistent distances) and our fit to the digitised MIGHTEE-HI points, both in the window (circles); the survey’s own fits to all radii under four mass-to-light conventions (squares); and SPARC in the window with mass-to-light ratios fitted to the rotation curves themselves (grey). The solid and dashed lines are $\kappa = 1/2$ against ρ_Λ and against ρ_{crit} ; the grey band is estimator A.

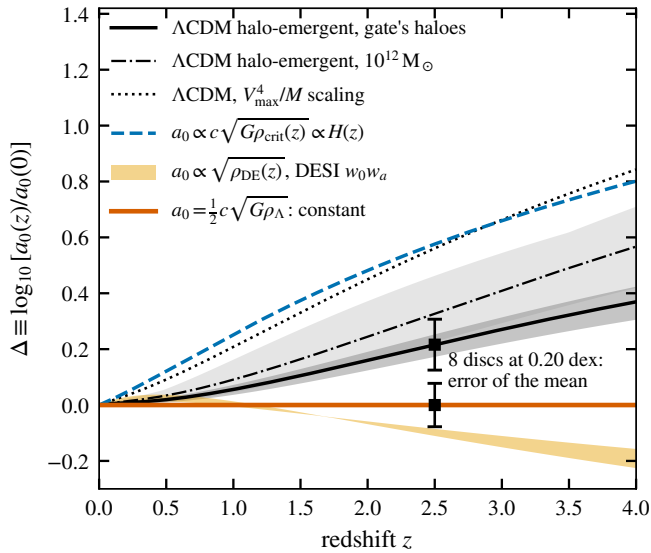


Figure 4. $\Delta(z)$ for the laws of Section 4.1: a scale anchored to Λ (thick horizontal line), a scale anchored to the critical density (dashed), and the Λ CDM halo-emergent scale for the haloes selected by the gate of Section 4.4 (solid black; dark band: the gate’s examples and $V_f / V_{200} = 1.0$ – 1.2), for $10^{12} M_\odot$ (dot-dashed), and its range over halo mass and concentration–mass relation (light band); dotted: the V_{max}^4 / M scaling. The narrow shaded band is the density mapping under the DESI DR2 $w_0 w_a$ fits, shown for scale. The two squares at $z = 2.5$ mark the decision values 0 and +0.22, with error bars equal to the standard error of the sample mean of the recommended design (eight galaxies at 0.20 dex) under each hypothesis.

Table 10. Error amplification of equation (25). The last column is the error on $\log_{10} \hat{a}_0$ for a 5 per cent velocity error and a 0.10 dex baryonic-mass error.

$y = g_{\text{bar}} / a_0$	n	A_{obs}	A_{bar}	$\sigma(\log_{10} \hat{a}_0)$
0.01	−0.475	2.1	1.1	0.14
0.1	−0.425	2.4	1.4	0.17
0.3	−0.376	2.7	1.7	0.20
1.0	−0.291	3.4	2.4	0.29
3.0	−0.186	5.4	4.4	0.50
6.4	−0.110	9.1	8.1	0.90

Unlike the deep-regime formula $g_{\text{obs}}^2 / g_{\text{bar}}$, this is unbiased at any acceleration. Its precision is another matter. Taking the logarithm of $v(y) = g_{\text{obs}} / g_{\text{bar}}$ and differentiating, with $n \equiv d \ln v / d \ln y$, gives $n d \ln y = d \ln g_{\text{obs}} - d \ln g_{\text{bar}}$; since $d \ln \hat{a}_0 = d \ln g_{\text{bar}} - d \ln y$,

$$d \ln \hat{a}_0 = -\frac{1}{n} d \ln g_{\text{obs}} + \left(1 + \frac{1}{n}\right) d \ln g_{\text{bar}}, \quad n = -\frac{\sqrt{y}/2}{e^{\sqrt{y}} - 1}. \quad (26)$$

Errors in the kinematics are therefore multiplied by $A_{\text{obs}} = 1/|n|$, and errors in the baryonic mass by $A_{\text{bar}} = A_{\text{obs}} - 1$. In the deep regime $n \rightarrow -1/2$, so $A_{\text{obs}} = 2$ and $A_{\text{bar}} = 1$: the familiar $\hat{a}_0 = g_{\text{obs}}^2 / g_{\text{bar}}$, in which a velocity error enters four times. Towards high acceleration $n \rightarrow 0$ and both factors diverge, because there the relation is $g_{\text{obs}} = g_{\text{bar}}$ and no longer contains a_0 (Table 10, Fig. 5).

This is why existing samples cannot decide between the laws. The star-forming galaxies with resolved kinematics at $z \sim 1$ – 2.5 are massive and compact. In the largest homogeneous sample, RC100 (Nestor Shachar et al. 2023), the median is $y = 1.45$, the central 68 per cent lie in $0.33 < y < 5.3$, and only 15 of 99 galaxies lie below $y = 0.3$. At the median, a baryonic-mass uncertainty of 0.2 dex, typical at these redshifts, becomes 0.6 dex in a_0 , which is larger than any of the separations in Table 9. Tully–Fisher zero points at high acceleration suffer from the same dilution, and in addition are degenerate

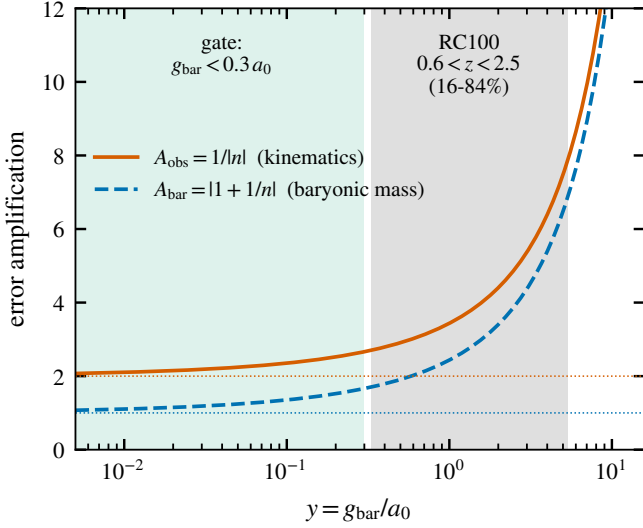


Figure 5. The factors by which errors in the kinematics (A_{obs} , solid) and in the baryonic mass (A_{bar} , dashed) are amplified when a_0 is inferred through equation (25). Dotted lines mark the deep limits, 2 and 1. The grey band is the central 68 per cent of the RC100 sample; the shaded region on the left is the gate proposed in Section 4.4.

with the evolution of disc sizes. The baryonic zero-point offsets of -0.44 and -0.27 dex at $z \approx 0.9$ and 2.3 found by Übler et al. (2017) therefore cannot be read as changes of a_0 , and the Tully–Fisher test of Limbach et al. (2008), at high acceleration and $z \leq 1.2$ where law (b) differs from constancy by at most 0.3 dex, is subject to the same limit.

4.3 What existing data say

A closed-form inversion of RC100. Nestor Shachar et al. (2023) tabulate, for 100 galaxies at $0.6 < z < 2.5$, the circular velocity at the effective radius and the dark-matter fraction inside it, $f_{\text{DM}} \equiv V_{\text{DM}}^2/V_c^2$. Both are outputs of their mass models, which use stellar mass-to-light ratios, gas masses from the scaling relations of Tacconi et al. (2018) and halo priors. We use their table 3 as posted in arXiv:2209.12199v1, the only arXiv version, in a transcription checked cell by cell; it has not been compared with the published table. By definition $g_{\text{bar}}/g_{\text{obs}} = 1 - f_{\text{DM}}$, and by equation (7) the same ratio is $1/\nu = 1 - e^{-\sqrt{y}}$. Therefore

$$e^{-\sqrt{y}} = f_{\text{DM}} \implies y = \left[\ln \frac{1}{f_{\text{DM}}} \right]^2 \implies \hat{a}_0 = \frac{(1 - f_{\text{DM}}) g_{\text{obs}}}{[\ln(1/f_{\text{DM}})]^2}. \quad (27)$$

The inversion adds no mass model to the authors’ own. It is defined for 99 galaxies (Fig. 6). The median is $\hat{a}_0 = 1.4 \times 10^{-10} \text{ m s}^{-2}$ with a scatter of 0.4 dex, as large as Table 10 leads one to expect. A straight-line fit gives

$$\frac{d \log_{10} \hat{a}_0}{dz} = -0.11 \pm 0.06. \quad (28)$$

Laws (b) and (c), fitted in the same way at RC100’s redshifts (the halo law for $10^{12} \text{ M}_{\odot}$, appropriate to these massive discs), have slopes of $+0.22$ and $+0.15$. The inversion itself does not manufacture a flat trend: dark fractions generated from equation (7) on the real redshifts and accelerations, with injected slopes of 0 , $+0.13$ and $+0.23$ and 0.05 of noise in f_{DM} , return those slopes to 0.01 .

The result is not a measurement, for five reasons, and we attach no

significance to the formal distance between the slope and either comparator. First, it restates the published decline of f_{DM} with redshift and inherits the inputs of the original fits. The dark fractions track the authors’ baryonic-mass prior: for the 41 galaxies with independently tabulated SED stellar and gas masses (Price et al. 2021), the residual from a constant a_0 correlates with the offset of RC100’s fitted baryonic mass from those masses (Spearman $\rho = +0.33$, $p = 0.036$). Secondly, RC100’s own table is not internally consistent in every row: two galaxies have $V_c^2 < 3.36\sigma_0^2$ at R_e , so that $V_{\text{rot}}^2 < 0$ by their equation 8, and 14 more lie below their cut $V_{\text{rot}}/\sigma_0 = 2.3$ at R_e ; dropping them moves the slope to -0.09 ± 0.06 , by 0.3σ . Thirdly, the sample is selected differently at different redshifts: controlling for the measured g_{obs} gives -0.14 ± 0.06 , and controlling for the inferred y gives $+0.01 \pm 0.05$, a control that also absorbs part of any real signal, since a change of a_0 changes y .

Fourthly, and decisively, the slope depends on the redshift calibration of the baryonic masses, which enters every f_{DM} . Suppose the baryonic masses carry a drift of β dex per unit redshift about the mean redshift, so that each g_{bar} is multiplied by $10^{\beta(z-\bar{z})}$. A drift of $\beta = -0.05$, a factor of 1.1 at either end of the range, moves the slope to $+0.02 \pm 0.06$ and erases the preference for constancy over the halo law, which then lies about 2σ away ($\beta = -0.075$ brings it within 1σ). For $\beta = -0.10$ the slope is $+0.16 \pm 0.07$, within 1σ of the $H(z)$ law and 2.3σ from constancy. One galaxy that sits exactly at the edge of the inversion, $f_{\text{DM}} = 0.02$, moves the slope by 0.4σ on its own. Fifthly, the pattern is not reproduced by an independent sample. For 117 galaxies of KMOS^{3D} with circular velocities from Übler et al. (2017), sizes from the survey catalogue (Wisnioski et al. 2019) and gas masses from scaling relations (93 of them not in RC100), every model leaves residuals that fall with redshift, the constant- a_0 model included. The reason is common to all models: at $z > 1.9$ between 27 and 54 per cent of these galaxies rotate more slowly than their own Newtonian baryons would require, which no model that adds gravity to the baryons can fit. Neither sample therefore decides between the laws of Table 9. Their limiting input is the baryonic mass, and a decisive sample needs gas masses measured galaxy by galaxy.

The MUSE lensed sample at $z \approx 1$. Two analyses of one survey point in different directions. Jeanneau et al. (2026) find that the baryonic Tully–Fisher zero point of 95 lensed galaxies at $z \approx 1$ has changed by 0.00 ± 0.06 dex with respect to $z = 0$; their atomic and molecular gas masses come from scaling relations and make up a median 70 per cent of M_b . Ciocan et al. (2026) fit the radial acceleration relation of 79 galaxies at $0.33 < z < 1.44$ and find $a_0(z \approx 1) = 2.38^{+0.12}_{-0.10} \times 10^{-10} \text{ m s}^{-2}$ (95 per cent interval), a rise faster than the $H(z)$ law: $\Delta \approx +0.30$ relative to the local $1.2 \times 10^{-10} \text{ m s}^{-2}$, and $+0.38$ relative to their own fitted $a_0(0) = 1.0 \times 10^{-10} \text{ m s}^{-2}$. Taken at face value this is the strongest published result against a constant a_0 , and we state it as such. It is not, however, independent of how the masses are obtained. The public per-galaxy products of the survey give, for each galaxy, the disc mass fitted jointly with a dark halo, which is the route of Ciocan et al. (2026), and an independent stellar mass from the spectral energy distribution (SED). Solving equation (7) for a_0 at the effective radius of 109 galaxies in three redshift thirds (Appendix B), the implied a_0 rises by $+0.57 \pm 0.09$ dex between the outer thirds with the fitted masses, against $+0.18$ for the $H(z)$ law. It is not recovered with SED stellar masses plus molecular gas (-0.11 ± 0.49 dex) or with SED stellar masses alone ($+0.04 \pm 0.14$ dex), but the data cannot say which masses are biased: the fitted minus SED stellar mass falls by 0.72 dex per unit redshift (95 per cent interval 0.50 – 0.97), and by 0.43 dex (0.15 – 0.72) at fixed SED stellar mass; restoring the rise on the SED route would need an

SED-mass bias of 1.8–2.2 dex per unit redshift, which is implausible; and the SED stellar masses plus molecular gas exceed the model’s own dynamical mass in many of the high-redshift galaxies. The rise is route-dependent: neither its nor constancy is established. At $z \approx 1$ the halo-emergent law cannot in any case be told from constancy (Table 9).

Outer rotation curves at $z \approx 1.5$. KURVS-CDFS (Puglisi et al. 2023) follows star-forming discs to about four effective radii. For the ten discs with tabulated dark-matter fractions the outermost points lie at $g_{\text{bar}}/a_0 = 0.06\text{--}0.67$ for the survey’s own gas fractions, the first public sample in the regime the test needs. At those radii the observed velocities must be corrected for pressure support, and the result turns on that correction and on the unmeasured gas. With the correction calibrated on simulations by Kretschmer et al. (2021) and the survey’s molecular-gas estimate, at a grid cell declared before this comparison (although after an earlier pressure model had been scored, so the choice of calibration was not blind; Appendix B), the outer accelerations exceed the constant- a_0 prediction by 3.3σ , while the rival’s central value sits at zero (-0.1σ). We report this against interest. It is not a detection. The outer velocities tabulated by the survey are its model curve evaluated at the last radius; with the measured outermost points the two numbers become $+2.4\sigma$ and -0.4σ . The lean disappears for a common velocity scale 4.5 per cent lower and for a pressure correction at 0.6 of the calibration ($+1.3\sigma$ and -2.0σ), and it reverses for a cold-gas mass of 1.5 times the stellar mass ($+1.2\sigma$ and -2.1σ). With gas and pressure support treated as nuisance parameters the likelihood ratio between the two laws is 0.5–1.7. What would settle it is the discs’ own cold gas and their outer pressure support, both measurable.

H I rotation curves at $z < 0.1$. Văršăteanu et al. (2026) fit the radial acceleration relation of 130 resolved H I-selected galaxies out to $z \approx 0.09$. Over that range $a_0 \propto H(z)$ changes a_0 by only 4.5 per cent, so the sample cannot separate the laws: its own linear trend, $a_1/a_0 = -1.04 \pm 1.51$ per unit redshift, is consistent with constancy (-0.7σ) and with $H(z)$ (-1.0σ). Anchored to SPARC at $z = 0$ the fitted trend becomes a formal 5σ rise; in their own table it is $(+5.23 \pm 1.05) \times 10^{-10} \text{ m s}^{-2}$ per unit redshift with mass-to-light ratios that vary between galaxies and $(-4.80 \pm 0.76) \times 10^{-10}$ with a constant ratio of 0.6. The authors find no significant evolution of the acceleration scale within their sample and attribute an apparent trend of the Tully–Fisher zero point (8.7σ in the forward fit, 3.4σ in their fiducial inverse fit) to selection; the earlier 19-galaxy analysis reported tentative evidence for evolution (Văršăteanu et al. 2025). In post hoc runs of a declared MIGHTEE-like scenario in our repository (SPARC-resampled mock samples, not the survey’s data; Appendix B), the formal error of the anchored trend was 5.2 times smaller than its scatter, and a constant a_0 gave a formal $z > 3$ in 11 to 32 per cent of the mocks. None of this is evidence for constancy or against it.

Discs at $z \approx 2.5\text{--}14$. Every further public sample at these redshifts that we could assemble has been scored under criteria fixed before the scoring (Zimmerman 2026b), and none measures a_0 . The discs with measured gas are massive and their last measured radii lie close to the Newtonian regime, where Table 10 multiplies every baryonic-mass error several-fold; the discs that reach low accelerations have no gas measurement; and in several the published baryons alone exceed the dynamics. Pools of the published objects at $z \geq 4$ with both stellar and gas masses do not separate the two laws once the common baryonic-mass calibration is uncertain by 0.30 dex, and pools with stellar masses only, which appear to separate them, lie above both predictions, in the direction the missing gas would remove. The repository analysis behind this paragraph fails one of its

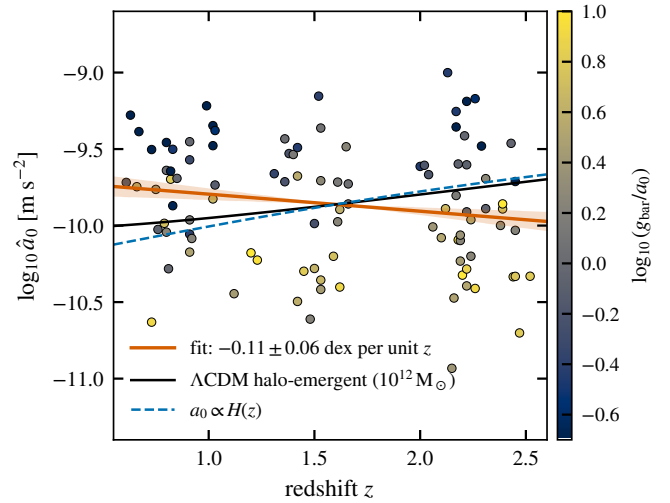


Figure 6. The acceleration scale inferred from the RC100 dark-matter fractions through equation (27), for 99 galaxies, coloured by the acceleration at which each is measured. The band is the fitted trend with its 1σ slope error. The thin solid and dashed lines are the halo-emergent ($10^{12} M_{\odot}$) and $H(z)$ laws, normalised at the mean redshift. The anticorrelation of $\hat{a}_0$ with y is the signature of noise in f_{DM} , which moves the two in opposite directions. The trend is conditional on the redshift calibration of the baryonic masses (Section 4.3): a drift of -0.05 dex per unit redshift moves it to $+0.02 \pm 0.06$.

own frozen controls (a recomputation of committed residuals agrees only to 2×10^{-3} dex against a 10^{-4} tolerance), so we cite it only qualitatively.

Rotation curves at $z \sim 2$. Milgrom (2017) showed that the outer rotation curves of the massive discs of Genzel et al. (2017) are compatible with the local a_0 and all but exclude a value about four times larger. Law (b) predicts a factor of 3.0 at $z = 2$; whether those data disfavour it would need the calibration analysis applied to RC100 above, and they are subject to the same limits: high accelerations and model-dependent baryonic masses.

Summary. The existing rotation curves at $z \approx 1\text{--}14$ decide nothing between the laws of Table 9. In every case the limiting input is the baryonic mass, and above all its common calibration. At $z \gtrsim 2$ the gas masses come from scaling relations (Tacconi et al. 2018, 2020) or from CO with a conversion factor whose metallicity dependence is uncertain (Bolatto et al. 2013). The dust-to-molecular-gas ratios of the ACE survey at $z \approx 2.2$ (Geesink et al. 2026), each source with its own conversions, lie 0.2–0.7 dex below the local relation of Bertemes et al. (2018) at the same metallicity, depending on how that relation is extrapolated in metallicity (Zimmerman 2026b). No calibration independent of the dynamics under test is known to reach 0.1 dex.

4.4 The measurement that decides

The gate. Table 10 shows that a galaxy is useful only if it is measured at low acceleration. We propose $g_{\text{bar}}(R_{\text{out}}) < 0.3a_0$, where $A_{\text{obs}} \leq 2.7$ and $A_{\text{bar}} \leq 1.7$. With $g_{\text{bar}} = \Gamma GM_b/R^2$, $\Gamma \approx 1.2$ and a_0 the working value of equation (4), this reads

$$M_b < 1.7 \times 10^8 M_{\odot} \left(\frac{R_{\text{out}}}{\text{kpc}} \right)^2, \quad (29)$$

for example $M_b = 5 \times 10^9 M_{\odot}$ traced to 6 kpc. Such galaxies rotate at $V_f \approx 83\text{--}125 \text{ km s}^{-1}$, which fixes the halo mass and so the decision

value of equation (23). At $z \simeq 2.5$ they can be resolved only with the help of gravitational lensing, which adds a requirement on the lens model: a magnification uncertainty below about 0.05 dex (see the budget below). Rotation must dominate ($V/\sigma > 1.5$ from a forward-modelled fit), the inclination should be between 35° and 75° , and the gas mass must be measured and not inferred from a scaling relation. A practical sequence is a JWST NIRSpec integral-field observation first, to establish rotation and bound the acceleration ($H\alpha$ and $[\text{O III}]$ fall together in the G235H grating for $2.3 < z < 3.8$), followed by ALMA CO(3–2), which falls in Band 3 for $2.0 < z < 3.1$ and gives the molecular gas mass and a second kinematic tracer. We know of no published object that passes every gate. In a ledger of candidates re-derived from the primary papers, the only object at $z \geq 2$ that could pass the acceleration gate is OLAS M0717-02064 (Hirtenstein et al. 2019): $z = 2.07$, magnification 6.5, $\log_{10} M_*/M_\odot = 8.1$, and rotation-supported with $v/\sigma = 1.6$. It lies just below the preferred redshift window, where Table 9 still separates the laws by 0.16 dex for the gate’s haloes, and it has not been observed with JWST. Its gas mass and the uncertainty of its lens model are not yet measured.

How many galaxies? Each galaxy that passes the gate gives $\Delta_i = \log_{10}[\hat{a}_0/a_0(0)]$. Under constancy Δ_i is drawn from a normal distribution of mean 0 and variance $s_0^2 = \sigma_m^2 + \sigma_{\text{int}}^2$, where σ_m is the measurement error and σ_{int} is the intrinsic scatter of the relation, about 0.034 dex in g_{obs} (Desmond 2023), which equation (26) amplifies to 0.09 dex in $\hat{a}_0$ at $y = 0.3$. Under the halo law the mean is $\Delta = +0.22$ and the variance $s_1^2 = s_0^2 + \sigma_h^2$, where $\sigma_h = 0.13$ dex is the halo-to-halo scatter of 0.11 dex in concentration (Dutton & Macciò 2014) propagated through equation (22) at the gate’s halo mass. For two exactly specified predictions with one Gaussian error σ , the expected logarithm of the likelihood ratio is $\Delta^2/2\sigma^2$, so expected odds of 20:1 (strong evidence on the scale of Kass & Raftery 1995) need $\sigma \leq \Delta/\sqrt{2\ln 20} = \Delta/2.45$, that is 0.088 dex for the halo law and 0.24 dex for $H(z)$. With both scatters the expected log-odds from N galaxies when constancy is true are

$$\langle \ln B \rangle = N \left[\ln \frac{s_1}{s_0} - \frac{1}{2} + \frac{s_0^2 + \Delta^2}{2s_1^2} \right], \quad (30)$$

and the expression with $s_0 \leftrightarrow s_1$ applies when the halo law is true. We call $\exp(\ln B)$, the smaller of the two, the expected odds. Expected odds are not the probability of reaching them: at the idealised limit of equation (30) with one pair of point hypotheses, $\ln B$ is normally distributed about $\ln 20$ and reaches 20:1 only half the time, pointing the wrong way 11 per cent of the time. We therefore also give, from a Monte Carlo of the exact likelihood ratio, the probability of reaching 20:1 under each hypothesis and of evidence pointing the wrong way ($\ln B < 0$). Table 11 gives the design. One galaxy at 0.13 dex yields 1.8:1. Expected odds of 20:1 need five galaxies at 0.10 or 0.13 dex, or eight at 0.20 dex, and at those N the chance of actually reaching 20:1 is only 0.53–0.68 under the less favourable hypothesis, with misleading evidence in up to 10 per cent of trials; reaching 20:1 with 90 per cent probability needs 9, 12 and 20 galaxies. The designs of an earlier deposited version of this test (Zimmerman 2026a) and of the previous draft of this paper, which used the rise of a $10^{12} M_\odot$ halo, are superseded: at the gate’s halo mass, two galaxies at 0.10 dex give 4.4:1 and four at 0.20 dex give 4.5:1.

The Λ CDM prediction is itself uncertain. We carry the choice of concentration–mass relation as a composite hypothesis: under the halo law, Δ itself is drawn once for the whole sample from a normal distribution centred on +0.22 with width σ_{sys} , a term that does not average down. The relations of Dutton & Macciò (2014) and Duffy et al. (2008) differ by 0.22 dex at the gate’s halo mass. With $\sigma_{\text{sys}} = 0.10$ dex, which allows the rise to be lower as well as higher, eight galax-

ies at 0.20 dex become 27; with 0.20 dex, 48 (Table 11). The relation spread, not the measurement error, is then the limiting term.

The error budget. Gravitational lensing conserves surface brightness, so $g_{\text{bar}} \propto M_b/R^2$ does not depend on the magnification μ , while $g_{\text{obs}} = V^2/R$ scales as $\mu^{1/2}$ through the radius. Equation (26) then gives

$$\sigma_m^2 = A_{\text{obs}}^2 \left[\left(\frac{2}{\ln 10} \frac{\delta V}{V} \right)^2 + \left(\frac{1}{2} \delta \log_{10} \mu \right)^2 \right] + A_{\text{bar}}^2 (\delta \log_{10} M_b)^2, \quad (31)$$

where $\delta \log_{10} M_b$ is the error of the mass scale at fixed magnification. At $y = 0.2$ ($A_{\text{obs}} = 2.5$, $A_{\text{bar}} = 1.5$), three budgets reproduce the rows of Table 11: $\sigma_m = 0.10$ dex needs $(\delta V/V, \delta \log_{10} M_b, \delta \log_{10} \mu) = (2.5 \text{ per cent}, 0.04, 0.04)$; 0.13 dex needs (3 per cent, 0.06, 0.05); and 0.20 dex needs (5 per cent, 0.10, 0.06). The first two are beyond present practice for the baryonic mass, since the conversion from CO luminosity to molecular mass is rarely known to 0.06 dex. The third is within reach. The realistic programme is therefore about eight galaxies at 0.20 dex for expected odds of 20:1, and about twenty for a 90 per cent chance of reaching them, provided that their mass errors are independent, which the next paragraph examines.

A common calibration does not average down. Part of $\delta \log_{10} M_b$ is shared by the whole sample: the conversion from CO luminosity to molecular mass, the dust-to-gas ratio and the zero point of the stellar masses. A shared error δ_c on $\log_{10} M_b$ adds the same $\sigma_C = A_{\text{bar}} \delta_c$ to every galaxy. The covariance of the N values of Δ_i is then $s^2 \mathbf{I} + \sigma_C^2 \mathbf{J}$, with $\mathbf{J}$ the matrix of ones and s^2 the per-galaxy variance of equation (30), and the expected log-odds are the Kullback–Leibler divergence between two N -dimensional normal distributions, which reduces to equation (30) when $\sigma_C = 0$. Along the direction of the mean the variance is $s^2/N + \sigma_C^2$, so the separation of the sample means cannot exceed Δ/σ_C however many galaxies are observed. Table 12 gives the result at $y = 0.2$, where $A_{\text{bar}} = 1.52$. A common error of 0.10 dex reduces eight galaxies at 0.20 dex from 20:1 to 2.5:1 against the halo law. Odds of 20:1 from the sample means require $\delta_c \leq 0.06$ dex against the halo law and $\delta_c \leq 0.15$ dex against the $H(z)$ law. The gas-calibration spread of Section 4.3, 0.2–0.7 dex at $z \approx 2.2$, is 0.09–0.48 dex in M_b for a gas fraction of one half. The deciding measurement therefore needs, besides the galaxies, a calibration of the gas mass at $z \approx 2.5$ that does not exist today.

Why the calibration must come from outside. In the deep limit g_{obs} depends on a common mass-calibration factor f and on a_0 only through their product $f a_0$, so a sample of deep galaxies measures little more than $f a_0$. The two can be separated by the rotation curves alone only through the curvature of equation (7), which needs points on both sides of the bend. With f free, the Fisher matrix gives $\sigma(\log_{10} a_0) \geq 3\sigma/\sqrt{N}$ for any design with N points of error σ ; for 20 points at 0.1 dex spread evenly in $\log y$, the estimate stays below 0.1 dex only if they run from $y = 0.001$ to at least $y \approx 17$, reaches 0.096 dex at best if they start at $y = 0.01$, and never reaches 0.1 dex if they start at $y = 0.1$. These statements are formalised, as implications of the assumed law, in a companion note (Zimmerman 2026b); we have recomputed the numbers for equation (7). There are thus two routes: rotation curves that span the bend widely enough to calibrate f themselves, which no curve at high redshift does (it would need about four decades in y), or an external calibration of f to better than 0.1 dex. The gate of this section selects deep galaxies and so takes the second route. This reconciles the design above, whose odds assume independent mass errors, with that note’s conclusion that in every existing test the limit that survives an unlimited sample is the calibration of the baryonic masses: the two agree once the shared part of the calibration is carried, as in Table 12.

Table 11. The design at the gated decision value ($\Delta = +0.22$ dex at $z = 2.5$), for N galaxies passing the gate, each with measurement error σ_m on $\log_{10} \hat{a}_0$, with intrinsic (0.09 dex) and halo-to-halo (0.13 dex) scatter and independent mass errors. N_{20} : the smallest N with expected odds $\exp(\ln B) \geq 20$, the smaller of the two cases (constancy true; halo law true). P_{20} : the probability of reaching 20:1 at N_{20} under the less favourable hypothesis. P_{wrong} : the probability that the evidence points the wrong way at N_{20} , under the more exposed hypothesis. N_{90} : the smallest N that reaches 20:1 with probability 0.9 under both hypotheses. Columns 6–9: the same with the Λ CDM prediction uncertain by a width σ_{sys} shared by the sample (Section 4.4). Probabilities from 2×10^4 – 10^5 Monte Carlo samples.

σ_m [dex]	N_{20}	$\sigma_{\text{sys}} = 0$			$\sigma_{\text{sys}} = 0.10$ dex		$\sigma_{\text{sys}} = 0.20$ dex	
		P_{20}	P_{wrong}	N_{90}	N_{20}	N_{90}	N_{20}	N_{90}
0.10	5	0.68	0.07	9	10	29	14	40
0.13	5	0.55	0.10	12	13	44	20	63
0.20	8	0.53	0.10	20	27	110	48	167

Table 12. Expected odds with a baryonic-mass calibration error δ_c shared by every galaxy, at $y = 0.2$ ($\sigma_C = A_{\text{bar}} \delta_c$, $A_{\text{bar}} = 1.52$), with the intrinsic and halo-to-halo scatter of Table 11. Columns 3–6: eight galaxies at $\sigma_m = 0.20$ dex (with P_{20} , the probability of reaching 20:1 under the less favourable hypothesis) and five at 0.10 dex against the gated halo law ($\Delta = 0.22$), and eight at 0.20 dex against the $H(z)$ law ($\Delta = 0.58$). Columns 7–8: Δ/σ_C , the largest separation of the sample means in standard deviations for any number of galaxies.

δ_c [dex]	σ_C [dex]	halo, 8@0.20 odds	P_{20}	halo, 5@0.10 odds	$H(z)$, 8@0.20 odds	halo $H(z)$ Δ/σ_C
0.00	0.000	20:1	0.53	40:1	$9.1 \times 10^{11}:1$	–
0.05	0.076	6.2:1	0.26	9.0:1	$1.3 \times 10^6:1$	2.8
0.10	0.152	2.5:1	0.04	3.2:1	290:1	1.4
0.15	0.228	1.7:1	0.00	2.2:1	17:1	0.9
0.20	0.304	1.5:1	0.00	1.9:1	5.4:1	0.7

The decision rule. The statistic is the mean of Δ_i over the galaxies that pass the gate, with $\hat{a}_0$ from equation (25) and $a_0(0)$ fixed beforehand from the local relation using the same mass-to-light conventions. A result near 0 supports constancy: law (a), and with it plain MOND. A result near +0.22 or above supports an emergent or $H(z)$ -coupled scale and excludes equation (1) with a constant ρ_Λ . A result inconsistent with all of them counts against all of them. No nuisance term may be introduced after the data are seen. Because $a_0(0)$ and $\hat{a}_0(z)$ are obtained with the same function and conventions, the test does not depend on κ or on H_0 , and it depends on the choice of transition function only weakly provided $a_0(0)$ is calibrated on local galaxies at matching accelerations. It does depend on the stellar and gas mass scales being on a common calibration at both epochs (Table 12).

5 WHAT THE RELATION DOES NOT EXPLAIN

The coefficient is not derived. We know of no derivation of $\kappa = 1/2$. The simplest physical identification, equating the Unruh temperature of a_0 with the temperature of the de Sitter horizon, gives $a_0 = cH_\Lambda$, which is six times too large; the form $cH_\Lambda/2\pi$ ($\kappa = 0.461$) is allowed by the data as readily as $1/2$ (Table 6), and on a common H_0 footing Milgrom’s $cH_0/2\pi$ fits SPARC at least as well as $\frac{1}{2}c\sqrt{G\rho_{\text{crit}}}$ (Section 3.9); $\kappa = 1/2$ is a fitted value. Nothing in the redshift test changes if κ lies anywhere in 0.4–0.6.

A scale is not a dynamics. Equation (1) fixes a number. It does not say whether the low-acceleration behaviour of galaxies is a modification of gravity, a modification of inertia, or a regularity of dark matter. It is therefore silent on gravitational lensing, the cosmic microwave background, the growth of structure, the external field effect (Chae et al. 2020) and the Solar System. Nor does it remove the need

for the cold matter whose density the microwave background and the growth of structure measure: that mass is still required. In particular, equation (7) is used here only to describe rotation curves; read as a modified-gravity law and extrapolated to Solar System accelerations it is constrained by planetary ephemerides (Hees et al. 2016).

Clusters of galaxies. Dynamics governed by a_0 leave a residual mass discrepancy of about a factor of two in rich clusters (Sanders & McGaugh 2002; Famaey & McGaugh 2012). Anchoring a_0 to Λ does not change that.

The mass-to-light ratio. The value $\kappa = 1/2$ with ρ_Λ is tenable only if $Y_{\text{disc}} \simeq 0.56$ at $3.6 \mu\text{m}$ on the Planck-consistent distances (0.62 on the tabulated distances; Table 2). An independent determination well outside 0.45–0.75 would exclude it. The ρ_{crit} reading needs $Y_{\text{disc}} \simeq 0.47$ (0.52). The deep regime shows how much is at stake: with the SED ratios of MIGHTEE-HI, a K_s -band median of 0.35, the coefficient is 0.90 ± 0.07 , and with a fixed $Y_K = 0.6$ it is 0.58 ± 0.05 (Section 3.8). The stellar mass calibration, not the number of galaxies, limits this test.

Constancy is shared. A constant a_0 at $z = 2.5$ would not distinguish equation (1) from a MOND constant of unknown origin. It would exclude the coupling to $H(z)$, which the coincidence $a_0 \sim cH_0$ has suggested since 1983, and it would be in tension with the halo-based expectation of Section 4.1, although the scale inferred for galaxies at fixed baryonic mass need not follow the central acceleration of a fixed-mass halo, so a constant result would constrain Λ CDM only through such a calibration.

Dark energy may not be a constant. If the evidence for evolving dark energy (DESI Collaboration et al. 2025) is confirmed, ρ_Λ in equation (1) will need redefinition. The redshift test is insensitive to this at the level of 0.1 dex (Section 4.1d).

6 CONCLUSIONS

(i) The only acceleration that can be formed from c , G and the density of the cosmological constant is $\kappa c\sqrt{G\rho_\Lambda}$. For $\kappa = 1/2$ it is $9.36 \times 10^{-11} \text{ m s}^{-2}$; the same expression with the critical density is $1.13 \times 10^{-10} \text{ m s}^{-2}$. Its equivalent forms are identities and carry no separate evidence.

(ii) On SPARC, with the Hubble-flow distances on the H_0 used for ρ_Λ , three estimators give $\kappa = 0.450 \pm 0.073$ (Tully–Fisher intercept with its finite-depth correction), 0.58 ± 0.18 (shape only, immune to a common distance error but fragile to the bulge mass-to-light ratio) and 0.43 ± 0.08 (profile likelihood with the mass-to-light ratio free per galaxy). The value $1/2$ is consistent with all three, as are $cH_\Lambda/2\pi$ and $cH_0/2\pi$; on either common footing Milgrom’s 2π form is at least as close to the data as $1/2$. Against that, the joint inference of Desmond (2023) puts $1/2$ at 2.6σ on the ρ_Λ footing. Each estimator’s value depends on the H_0 convention by up to 0.1, and injection tests show

that each returns a known a_0 . The thermal identifications cH_Λ and $2cH_\Lambda$ are excluded.

(iii) In the deep regime, $g_{\text{bar}} < 0.2a_0$, the per-galaxy slope of SPARC and MIGHTEE-HI agrees with the slope of equation (7) at the sampled accelerations, within 0.1 – 1.6σ , and excludes a slope of 0.75 at more than 4σ . The amplitude follows the stellar mass-to-light convention: $\kappa_\Lambda = 0.90 \pm 0.07$ with the SED ratios of MIGHTEE-HI and 0.58 ± 0.05 with its fixed $Y_K = 0.6$. Mass-to-light ratios fitted to the rotation curves without the relation lower the deep-regime value by a factor of 1.7 to 2.2 and must not be used.

(iv) Because stellar and gas masses sum to the baryonic mass, the two calibrations trade against each other and leave a floor $s_Y s_g / (s_Y^2 + s_g^2)^{1/2} = 9.5$ per cent on a_0 that no sample size removes. Separating $1/2$ from 0.461 needs 2.6 per cent. At fixed Ω_Λ only the product κH_0 enters the prediction, so κ cannot be quoted without H_0 .

(v) The Λ anchor predicts a constant a_0 . A critical-density anchor predicts $+0.58$ dex at $z = 2.5$. Λ CDM haloes predict a rise that depends on halo mass; for the haloes of discs that can be measured at low acceleration it is $+0.22$ dex, with a model systematic of about 0.2 dex from the concentration–mass relation. Inferring a_0 from one galaxy amplifies kinematic errors by $1/|n|$ and mass errors by $1/|n| - 1$, factors of 4 and 3 at the accelerations of existing high-redshift samples, which is why those samples do not decide the matter.

(vi) A closed-form inversion of the 99 usable RC100 galaxies gives $d \log_{10} a_0 / dz = -0.11 \pm 0.06$, but the dark fractions are model outputs that track the authors’ baryonic-mass prior, a drift of -0.05 dex per unit redshift in the baryonic-mass calibration erases the trend, and an independent KMOS^{3D} sample does not reproduce it. The MUSE-DARK rise of a_0 at $z \approx 1$, the strongest published result against constancy, is not recovered with SED stellar masses plus molecular gas, and the data cannot say which masses are biased. The lean towards $a_0 \propto H(z)$ in KURVS at $z \approx 1.5$ rests on the pressure correction, the gas and the velocity scale. MIGHTEE-HI/LADUMA cannot separate the laws, and no sample at $z \approx 2.5$ – 14 measures a_0 .

(vii) About eight lensed rotators at $z \approx 2.5$ with $g_{\text{bar}} < 0.3a_0$, each measured to 0.20 dex in a_0 (five at 0.10 dex), give expected odds of $20:1$ between constancy and the halo-emergent rise once intrinsic and halo-to-halo scatter are included; reaching $20:1$ with 90 per cent probability needs about twenty (nine), and carrying the concentration–mass relation as a shared systematic of 0.1 dex raises these numbers by factors of two to five. All of this assumes a baryonic-mass calibration shared by the sample to 0.06 dex (0.15 dex against the $H(z)$ law), which cannot come from the rotation curves themselves; the gas prescriptions in use at $z \approx 2.2$ differ by 0.2 – 0.7 dex. A calibrated gas mass is the input still missing.

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for it. The ordering of the observing sequence in Section 4.4 was first suggested in an exchange with an OpenAI model. This work made use of the SPARC database (Lelli et al. 2016), of the tables of Nestor Shachar et al. (2023), Price et al. (2021) and Puglisi et al. (2023), of the public per-galaxy products of the MUSE-DARK survey (Ciocan et al. 2026), and of the published fits of Văršeteanu et al. (2026). The author received no funding for this work and declares no conflict of interest.

DATA AVAILABILITY

All scripts and outputs are in the public repository <https://github.com/carlzimmerman/zimmerman-formula>, in the directory `qwen_claude_field_theory/papers_2026/mnas_submission_2026_v3` (Appendix C); the version used for this paper is the tagged release `mnas-v3.1`. The SPARC data are available at <http://astroweb.cwru.edu/SPARC/>. The RC100 table is from Nestor Shachar et al. (2023); the transcription used here (`real_research/data/rc100_nestorshachar2023_table3_CORRECTED.csv`) was checked against table 3 of arXiv:2209.12199v1 and has not been compared with the published table. An earlier transcription in our repository had 17 wrong cells, three of which enter the inversion; correcting them moves the RC100 slope by less than 0.001 dex per unit redshift. The MIGHTEE-HI points were digitised from Văršeteanu et al. (2025), and the digitised values are in the repository. The published tables behind the high-redshift analyses (22 sources, including RC100, the KMOS^{3D} release, MUSE-DARK and MIGHTEE-HI/LADUMA, but not the KURVS tables) were audited for transcription and internal consistency: 3703 sampled cells in 29 tables showed no transcription mismatch, no erratum is registered for any of the published source papers, and no error found in a source table changes a result quoted here. Like RC100, two further tables behind the $z \gtrsim 2.5$ census (Förster Schreiber et al. 2018; Umehata et al. 2026) were checked against their arXiv v1 versions only. Earlier versions of parts of this work are deposited at Zenodo (Zimmerman 2026a,b).

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APPENDIX A: THE ACCELERATION SCALE OF AN NFW HALO

The NFW density profile is $\rho(r) = \rho_s/[x(1+x)^2]$ with $x = r/r_s$. Integrating, the mass inside r is

$$M(<r) = 4\pi\rho_s r_s^3 f(x), \quad f(x) = \ln(1+x) - \frac{x}{1+x}. \quad (\text{A1})$$

The halo is bounded at r_{200} , inside which the mean density is $200\rho_{\text{crit}}(z)$, and the concentration is $c_{200} = r_{200}/r_s$. Dividing $M(<r)$ by $M_{200} = M(<r_{200})$ removes ρ_s :

$$M(<r) = M_{200} \frac{f(x)}{f(c_{200})}, \quad (\text{A2})$$

$$g(r) = \frac{GM(<r)}{r^2} = \frac{GM_{200}}{f(c_{200})r_s^2} \frac{f(x)}{x^2}. \quad (\text{A3})$$

For small x , $f(x) = x^2/2 - 2x^3/3 + \dots$, so $f(x)/x^2 \rightarrow 1/2$. The acceleration is therefore largest at the centre, and with $r_s = r_{200}/c_{200}$

$$g_{\text{max}} = \frac{GM_{200} c_{200}^2}{2f(c_{200})r_{200}^2}. \quad (\text{A4})$$

The radius follows from the definition of M_{200} and from $\rho_{\text{crit}}(z) = 3H^2/8\pi G$:

$$M_{200} = \frac{4\pi}{3} 200 \frac{3H^2}{8\pi G} r_{200}^3 = \frac{100H^2 r_{200}^3}{G}, \quad (\text{A5})$$

$$r_{200} = \left(\frac{GM_{200}}{100H^2} \right)^{1/3}. \quad (\text{A6})$$

Hence $GM_{200}/r_{200}^2 = 100^{2/3}(GM_{200})^{1/3}H^{4/3}$, and substituting in equation (A4) gives equation (21). At fixed M_{200} the ratio of g_{max} at redshift z to its value today is equation (22). The virial velocity is $V_{200} = (10GM_{200}H)^{1/3}$, so the halo of a disc of flat velocity V_f has $M_{200} = (V_f/r)^3/(10GH)$ with $r = V_f/V_{200}$; this is how the gate's halo masses of Section 4.1 are obtained.

Numbers. For $M_{200} = 10^{12} M_\odot$ the relation of Dutton & Macciò (2014), $\log_{10} c_{200} = a + b \log_{10}(M_{200}/10^{12} h^{-1} M_\odot)$ with $a = 0.520 + 0.385 e^{-0.617z^{1.21}}$ and $b = -0.101 + 0.026z$, gives $c_{200} = 8.36$ and $r_{200} = 212$ kpc at $z = 0$, so $g_{\text{max}} = 8.1 \times 10^{-11} \text{ m s}^{-2}$. At $z = 2.5$ it gives $c_{200} = 3.85$, $r_{200} = 87$ kpc and $g_{\text{max}} = 1.72 \times 10^{-10} \text{ m s}^{-2}$. The ratio is 2.13. Term by term: $E^{4/3} = 3.767^{4/3} = 5.86$, and $[c^2/f(c)]_{2.5}/[c^2/f(c)]_0 = 18.9/52.0 = 0.363$.

A second scaling. The maximum circular velocity of an NFW halo is $V_{\text{max}}^2 = 0.216 V_{200}^2 c_{200}/f(c_{200})$. The Tully–Fisher-like ratio V_{max}^4/GM_b at a fixed ratio M_b/M_{200} therefore scales as $E^{4/3}[c/f(c)]^2$. Between $z = 0$ and 2.5 this is a factor 3.64, or +0.56 dex for $10^{12} M_\odot$; it is lower by the logarithm of any increase in M_b/M_{200} .

APPENDIX B: METHODS OF THE REPOSITORY RE-ANALYSES

The analyses of Section 4.3 quoted from our repository were run in our repository under criteria committed before each was scored; `paper_numbers.py` reads their committed outputs and confirms that the text quotes them (such checks carry no evidence). Two use the repository's monotone interpolating function, which equals equation (7) to 10^{-6} for $y \leq 1$ and differs from it by at most 2.4 per cent for $1 < y \leq 100$.

MUSE-DARK by baryon route. Inputs: the public per-galaxy products of the survey (one forward-modelled disc–halo fit per galaxy,

with a DC14 halo profile) and the SED stellar mass, for the 109 galaxies with both and no bulge. At the fitted effective radius R_e , $g_{\text{obs}} = v^2/R_e$ from the model’s projected velocity with the asymmetric-drift term added (the reading best supported by an inclination test). Three routes give g_{bar} : (i) $(1 - f_{\text{DM}})g_{\text{obs}}$ with the fit’s own dark fraction; (ii) a thin exponential disc of the SED stellar mass plus molecular gas from a Tacconi-type scaling relation and the fit’s (prior-dominated) H I surface density; (iii) as (ii) without the molecular gas. The sample is split into redshift thirds (37, 36 and 36 galaxies, mean $z = 0.52$, 0.88 and 1.20). In each third the implied scale s_* is the factor on a_0 that sets the median over galaxies of $\log_{10}[D_i/v(g_{\text{bar},i}/s_*a_0)]$ to zero, with $D_i = g_{\text{obs},i}/g_{\text{bar},i}$; the quoted change is $\log_{10} s_*$ of the third third minus that of the first, with the standard deviation of 10 000 galaxy-bootstrap resamples as its error. The mass drift is the slope of $\log_{10} M_{\text{fit}} - \log_{10} M_{*,\text{SED}}$ against z with its bootstrap 95 per cent interval, and at fixed SED mass with $\log_{10} M_{*,\text{SED}}$ as a covariate. A planted change of 0.30 dex in a_0 moves every level by that amount to within 0.04 dex.

KURVS. Inputs: the published redshifts, stellar masses, effective radii, velocity dispersions σ_0 and inclination-corrected velocities at the last measured radius R_{max} of the ten rotation-supported discs with tabulated dark fractions (Puglisi et al. 2023). g_{bar} is computed for thin exponential stellar and gas discs (gas scale twice the stellar) with the gas mass a declared multiple $\mu = M_{\text{gas}}/M_*$ of the stellar mass ($\mu = 0.25$ –4; the survey’s 40 per cent molecular fraction is $\mu = 0.67$). The circular velocity is $V_c^2 = V^2 + \alpha\sigma_0^2$, with $\alpha(R/R_e - 1)$ from the simulation calibration of Kretschmer et al. (2021). The statistic is the inverse-variance mean over the ten discs of $\log_{10}(g_{\text{obs}}/g_{\text{pred}})$ for each law, minus the same statistic for 75 SPARC galaxies processed identically (which removes the pipeline’s own offset at $z = 0$); per-object errors come from the velocity and σ_0 errors, inflated by $(\chi^2/\text{d.o.f.})^{1/2}$ when that exceeds one. The quoted σ is this statistic over its error; the decision cell is $\mu = 0.67$ with the published stellar masses. The likelihood ratio between the laws treats the gas fraction and the pressure correction as nuisance parameters over their declared ranges.

MIGHTEE-HI/LADUMA mocks. Post hoc, not the survey’s data: 130 SPARC-resampled galaxies at $0.02 < z < 0.09$ and an anchor library of SPARC-resampled fits at $z = 0$, generated under a constant a_0 with declared levels of shared systematics; the anchored slope is refitted in each mock and its formal error compared with its scatter over mocks.

APPENDIX C: REPRODUCIBILITY

The scripts below produce every number in the paper. Each prints its checks as PASS or FAIL and exits with a non-zero status on any failure; `reproduce_all.sh` runs them in sequence. It rewrites only the paper’s own products in its directory (`paper_numbers.out` and `.json`, the figures and the PDF) and runs the KMOS^{3D} lane in scratch copies, so no file outside the directory is changed. The checks in `paper_numbers.py` are labelled by kind, because not every check is evidence. Of its 85 checks, 39 are identities: they verify algebra or arithmetic on stated inputs, confirm that a table in the text equals the computed one, reproduce another script’s output, or confirm that the text quotes a committed output of a separate analysis correctly; they can fail only through a coding or transcription error. Thirteen evaluate published models at stated inputs. Twenty-eight can fail on the data. Five are injection tests (Sections 3.7 and 3.8), which feed an estimator synthetic data with a known answer. Only the last two kinds carry evidence. For the results of Section 4.3 that

are quoted from separate analyses, the evidence lies in those analyses (Appendix B). Paths are relative to the repository root, `v3/` stands for the directory named in the Data Availability statement, and `cfg/` for `campaign_fresh_gravity`.

- `v3/paper_numbers.py`: every table except Table 3; estimators B and C, the convention box, the design and its Monte Carlo; Sections 3.4, 3.8, 3.9, 3.10, 4.3 and 4.4; Appendix A. It runs the RC100 inversion on the corrected and on the earlier transcription and prints every number the correction moves.
- `v3/make_figures.py`: Figs 1–6, built from the functions of the script above.
- `real_research/reviews/mi_btfr_intercept_kappa_door_2026.py`: estimator A and Table 3; `real_research/reviews/kappa_h0_convention_audit_2026.py`: its Hubble-flow weight and its four H_0 conventions.
- `real_research/reviews/mi_distance_free_gbar_estimator_sparc_2026.py`: the earlier form of estimator B, reproduced as a check; `real_research/reviews/mi_a0_profile_likelihood_milgrom_footing_2026.py`: the earlier profile likelihood, reproduced exactly by estimator C’s code.
- `real_research/dark_sector_2026/L332_kmos3d_trend_replication.py`: the KMOS^{3D} comparison of Section 4.3; `cfg/CFG217_rc100_attack/`: the prior-tracking of the RC100 dark fractions.
- `real_research/reviews/deep_a0eff_audit_2026_09_25/`: the audit that traced the fitted mass-to-light ratios of Section 3.8.
- `cfg/CFG289_rc100_csv_bound/cfg289_build_corrected.py`: builds the corrected RC100 transcription from the earlier one and from the values read from the paper.
- `cfg/CFG262_musedark_zthirds_by_route/` and `cfg/CFG236_musedark_referee/`: MUSE-DARK by baryon route.
- `cfg/CFG140_kurvs_a0z.py`, `cfg/CFG160_kurvs_kretschmer.py`, `cfg/CFG165_kurvs_referee/`, `cfg/CFG189_kurvs_measured_markers/` and `cfg/CFG194_measured_markers_referee/`: KURVS.
- `cfg/CFG279_mightee_published_values/` and `cfg/CFG258_mightee_a0z_preflight/`: MIGHTEE-HI/LADUMA.
- `cfg/CFG269_highest_z_dynamics/`: the pools at $z \geq 4$ (its committed run fails its control C1; cited qualitatively).
- `cfg/CFG224_gas_calibration/` and `cfg/CFG238_gas_calibration_referee/`: the ACE comparison at $z \approx 2.2$.
- `cfg/CFG240_calibration_wall/`: the conditioning of a_0 with the mass scale free, with its formal statements.
- `cfg/CFG287_source_table_integrity/`: the audit of the source tables.

This paper has been typeset from a $\text{\LaTeX}$ file prepared by the author.